System:	**ABS 9.x** **ESP® 9.x** **ESP®plus 9.x**	**Equipment:**	**Bleeding device**
		Diagnostic interface:	**K-Line (opt.); CAN; FlexRay (customer specific)**
Brake circuit configuration:	**Diagonal (X)** **Front / Rear (II)**		
Customer:	**General**	**Diagnostic protocol:**	**Customer specific**

"D"	
Distribution:	Customer

History list:

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Department:	**Date:**	**Signature:**
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Modifications:

Version:	Modifications:
00	New document

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1 General

1.1 Acronyms

ABS	antilock braking system
ABS-SILA	ABS warning light
AC	Accumulator chamber
AV	outlet valve
ALS	longitudinal acceleration sensor
AYS	lateral acceleration sensor
BLS	brake light switch
BS	Brake switch
DTC	diagnostic trouble code
EBD	electronic brake distribution
ECU	electronic control unit
ESP	electronic stability program
ESP-MFL	ESP multifunction (warning) light
EV	inlet valve
FA	front axle
FCM	fault code memory
FL	front left
FR	front right
HAS	handbrake switch (park brake switch)
HSV	high pressure switching valve (priming valve)
MCC	master cylinder circuit
PATA	passive taster (TCS/DSC passive taster)
PB	process byte
PS (DG, DS)	pressure sensor
RA	rear axle
RFP	recirculation pump
RL	rear left
RR	rear right
SAS (LWS)	steering angle sensor
SV	solenoid valve
UMR	pump motor feedback
USV	pilot valve
UVR	Valve relay feedback
WSS	wheel speed sensor
YRS	yaw rate sensor

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2 General

This document describes each step for a repair bleeding process of an ABS or ESP system. This procedure respectively repair bleeding process has to carry out in case of the vacuum process is failed or a dry ABS/ESP hydraulic unit is assembled in filled vehicle.

2.1 Scope

All the Bosch ABS and ESP components, such as hydraulic modulator are delivered to the customer assembly plants in a dry state (free of brake fluid), that means that the primary and secondary circuits of the hydraulic modulator are complete dry.
To bleed respectively fill the secondary circuits of hydraulic modulator different valves must be energized.
At vehicles which fail the online evacuation and filling process or in case of replacement of the hydraulic modulator by a dry ABS / ESP unit a special repair bleeding procedure has to be carry out. During execution of this repair bleeding procedure different valves and pump motor have to be activated in an exact defined sequence and the operator has to open the bleeding screw at the respective wheel.
During the repair bleed process the brake fluid is pressed by the pressure of the bleeding device through the brake fluid reservoir, master cylinder, the entire brake system and the respective opened bleeding screw. In addition the brake pedal has to be actuated to support the bleeding process.

The execution of all required valve and pump motor activations in the defined sequence is realised by a diagnostic tester. This tester is connected with the ABS / ESP ECU via the diagnostic interface (OBD2) within the passenger cell (precondition: all components such as wiring harness and battery complete assembled).

The bleeding sequence is done in the same way as at a vehicle with a standard brake system without an ABS or ESP unit with exception of the additional time which is required to bleed the secondary and priming (only ESP) circuits of the ABS / ESP hydraulic unit.

The used bleeding device has to fulfil the requirements which are specified in this document.

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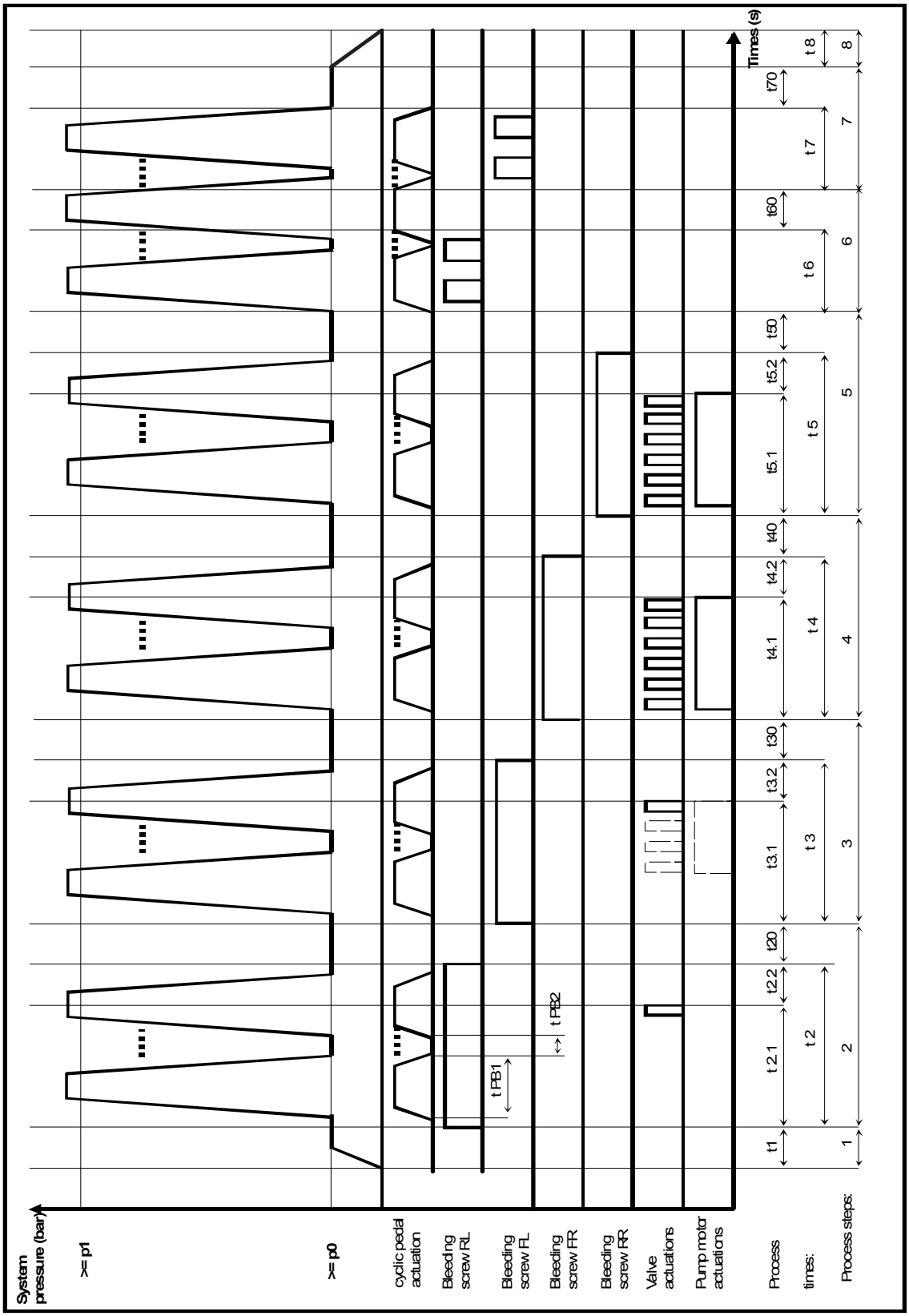
2.2 Process sequence

Diagram of the process sequence:



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Explanations to the process sequence diagram:

Process step:	Time:		Process description:
1	t1		At the beginning of the process the bleeding device and the electric connections have to be adapted. To guarantee a good bleeding result, a defined **absolute pressure p_0** at the bleeding device must be adjusted and the brake pedal must be cyclic actuated during the whole bleeding process. By the **cyclic brake actuation** is produced an appropriate **system pressure p_1** which supports the bleeding process considerably. In dependence of the accessibility of the bleeding screws the vehicle must be raised if necessary and all wheels be dismounted. The required **time t1** for the **handling 1** is dependent on the operator respectively the conditions at the vehicle.
2	t2	t2.1	The whole **repair bleeding process for the wheel RL** for the **bleeding time t_2** consists of the **phase duration $t_{2.1}$** and **waiting period $t_{2.2}$**. After starting the associated bleeding time t_x the appropriate wheel circuit is bleed. In parallel the associated valve controls (phase duration $t_{x.x}$) is started. During this time the specified valve activations are executed for the appropriate wheel to bleed the secondary circuit in the ABS / ESP hydraulic unit. After the specified valve controls (phase duration $t_{x.x}$) are finished, the appropriate wheel must be bleed as long as the brake fluid is seen in the bleeding tube (connected to the bleeding screw) free of air and bubbles. This required waiting period $t_{x.x}$ depends on the vehicle, the design and size of the base brake system. This time is to be specified by the customer.
		t2.2	
	t20		The **handling time t_{20}** is a time dependent on the operator and it is required to connect the bleeding bottle respectively opening and closing of the bleeding screws.
3	t3	t3.1	The whole **repair bleeding process for the wheel FL** for the **bleeding time t_3** consists of the **phase duration $t_{3.1}$** and **waiting period $t_{3.2}$**. For more details see the description of the process step 2.
		t3.2	
	t30		The **handling time t_{30}** is a time dependent on the operator. For more details see the description of the process step 2.
4	t4	t4.1	The whole **repair bleeding process for the wheel FR** for the **bleeding time t_4** consists of the **phase duration $t_{4.1}$** and **waiting period $t_{4.2}$**. For more details see the description of the process step 2.
		t4.2	
	t40		The **handling time t_{40}** is a time dependent on the operator. For more details see the description of the process step 2.
5	t5	t5.1	The whole **repair bleeding process for the wheel RR** for the **bleeding**

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		t5.2	**time t_5** consists of the **phase duration $t_{5.1}$** and **waiting period $t_{5.2}$**. For more details see the description of the process step 2.
	t50	The **handling time t_{50}** is a time dependent on the operator. For more details see the description of the process step 2.	
6	**t6**	In dependence of the brake split configuration and the used orifices in the ABS / ESP hydraulic unit the **wheel RL** must be bleed in the individual case again for the **bleeding time t_6 without valve controls**. **This step is optional.** The necessity of this additional bleeding step must be determined by attempts executed with vehicles with completely dry brake system. The responsibility for these trials has the customer.	
	t60	The **handling time t_{60}** is a time dependent on the operator. For more details see the description of the process step 2.	
7	**t7**	In dependence of the brake split configuration and the used orifices in the ABS / ESP hydraulic unit the **wheel FL** must be bleed in the individual case again for the **bleeding time t_7 without valve controls**. **This step is optional.** The necessity of this additional bleeding step must be determined by attempts executed with vehicles with completely dry brake system. The responsibility for these trials has the customer.	
	t70	The **handling time t_{70}** is a time dependent on the operator. For more details see the description of the process step 2.	
8	**t8**	At the end of the process the bleeding device and the electric connections have to be disconnected respectively the pedal stroking have to be stopped. The required **time t8** for the **handling 2** is dependent on the operator respectively the conditions at the vehicle.	

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3 Description of each process step

3.1 Process step 1: Handling 1

Handling 1		
Operator:		
☞ **Adapt the bleeding device with the adjusted absolute pressure p_0 at the brake fluid reservoir** ☞ **Adapt the electrical connections via the diagnostic interface (OBD2) in the vehicle** ☞ **Switch the ignition on** ☞ **Check the accessibility of the bleeding screws, if necessary raise the vehicle and dismount the wheels**		
Process step:	**Parameter:**	**Physical values:**
1	**Absolute pressure p_0**	**≥ y bar**
	Handling time t_1	**x s**

Notes:

In the plant rework area the vehicles are completely assembled. The electrical connection respectively the diagnostics are done via the OBD2 connector within the passenger cell. The operator has to switch on the ignition.

To support the bleeding process a bleeding device with the adjusted absolute pressure p_0 has to be adapted at the brake fluid reservoir.

The bleeding sequence of the wheels has not been changed in comparison to the Bosch generation 8. Following bleeding sequence has to be guaranteed; otherwise residual air can remain back in the brake system:

1. **Rear left (RL)**
2. **Front left (FL)**
3. **Front right (FR)**
4. **Rear right (RR)**
5. **Rear left (RL)– optional**

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6. Front left (FL) - optional

The necessity to execute these optional process steps 5 & 6 are dependent on following conditions:

- Brake split configuration
- Used orifices in the ABS/ESP hydraulic unit
- Filling state of the base brake system (partially or completely filled)

Regarding the bleed ability of a complete brake system, vehicles with a front/rear brake split configuration and an ABS/ESP system with small inlet orifices on the rear axle wheels are the worst case.

To find out if the optional bleeding steps 5 & 6 are really necessary, corresponding trials have to be executed by the customer.

The with x marked **handling time t_1** depends on the installation position of the brake fluid reservoir, the position of the diagnosis interface OBD2 in the vehicle interior and the conditions at the vehicle. The handling time t_1 must be determined together with the operators by trials. These trials are in responsibility of the customer and have to be executed by them self.

All with y marked **absolute pressure p_0** is to be inferred from the chapter **Process data (pressure and time values)**.

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3.2 Process step 2: Bleeding of the wheel circuit rear left (RL)

Bleeding of the wheel circuit rear left (RL)	
Operator:	**Diagnostic unit / ECU:**
☞ **Connect the bleeding bottle and open the bleeding screw at the wheel RL.** ☞ **Actuate the brake pedal alternately with a normal speed (actuated for the time t_{BP1}; released for the time t_{BP2}). This step could also automated by a pneumatic or electrical pedal actuation tool.** ☞ **Start the diagnostics (valve activations, etc.) via the tester and wait till the end message is displayed.**	**Specified valve activations plus the defined waiting time for the wheel RL are executed.**
☞ **Close the bleeding screw at the wheel RL and disconnect the bleeding bottle.** ☞ **Stop the brake pedal actuation.**	

Process step:	Parameter:		Physical values:
2	**Bleeding time t_2** (consist of the diagnostic times):	phase duration $t_{2.1}$	**y s**
		waiting period $t_{2.2}$	**x s**
	System pressure p_1		**y bar**
	Time t_{BP1}		**y s**
	Time t_{BP2}		**y s**
	Handling time t_{20}		**x s**

Notes:

The **bleeding screw RL** is opened all other bleeding screws are closed. The brake fluid is pushed out by the pressure of the bleeding device through the ABS/ESP hydraulic unit, the brake pipe, wheel brake cylinder respectively calliper and the bleeding screw. To support this process **the brake pedal** has been **alternate actuated** (actuated for the **time t_{BP1}**; released for the **time t_{BP2}**). This brake pedal stroking creates the **system pressure p_1**, which supports the bleeding process significantly. Analogously the **specified valve activations** for the **wheel RL** are executed for the **phase duration $t_{2.1}$** to bleed the secondary circuit of the ABS/ESP hydraulic unit.

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After the specified valve activations are finished, the wheel RL must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. The therefore needed additional **waiting period $t_{2.2}$** depends on the vehicle respectively on the design of the base brake system. This time must be specified by the customer. The total **bleeding time t_2** for the wheel RL consists of the phase duration $t_{2.1}$ and the waiting period $t_{2.2}$. The **handling time t_{20}** depends on the operator and is required to connect the bleeding bottle respectively to open and close the bleeding screw.

The with x marked **timing values** depend on the size of the brake system, the adjusted bleeding pressure respectively the conditions at the vehicle (e.g. accessibility of the bleeding screws, etc). These times must be determined by trials executed at the customer assembly plant. The handling time must be determined together with the operators. These trials are in responsibility of the customer and have to be executed by them self.

All with y marked **pressure and timing values** are to be inferred from the chapter **Process data (pressure and time values)** and **Diagnostic data (timing values)**.

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3.3 Process step 3: Bleeding of the wheel circuit front left (FL)

Bleeding of the wheel circuit front left (FL)	
Operator:	**Diagnostic unit / ECU:**
☞ **Connect the bleeding bottle and open the bleeding screw at the wheel FL.** ☞ **Actuate the brake pedal alternately with a normal speed (actuated for the time t_{BP1}; released for the time t_{BP2}). This step could also automated by a pneumatic or electrical pedal actuation tool.** ☞ **Start the diagnostics (valve activations, etc.) via the tester and wait till the end message is displayed.**	**Specified valve activations plus the defined waiting time for the wheel FL are executed.**
☞ **Close the bleeding screw at the wheel FL and disconnect the bleeding bottle.** ☞ **Stop the brake pedal actuation.**	

Process step:	**Parameter:**		**Physical values:**
3	**Bleeding time t_3** (consist of the diagnostic times):	phase duration $t_{3.1}$	**y s**
		waiting period $t_{3.2}$	**x s**
	System pressure p_1		**y bar**
	Time t_{BP1}		**y s**
	Time t_{BP2}		**y s**
	Handling time t_{30}		**x s**

Notes:

The **bleeding screw FL** is opened all other bleeding screws are closed. The brake fluid is pushed out by the pressure of the bleeding device through the ABS/ESP hydraulic unit, the brake pipe, wheel brake cylinder respectively calliper and the bleeding screw. To support this process **the brake pedal** has been **alternate actuated** (actuated for the **time t_{BP1}**; released for the **time t_{BP2}**). This brake pedal stroking creates the **system pressure p_1**, which supports the bleeding process significantly. Analogously the **specified valve activations** for the **wheel FL** are executed for the **phase duration $t_{3.1}$** to bleed the secondary circuit of the ABS/ESP hydraulic unit.

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After the specified valve activations are finished, the wheel FL must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. The therefore needed additional **waiting period $t_{3.2}$** depends on the vehicle respectively on the design of the base brake system. This time must be specified by the customer. The total **bleeding time t_3** for the wheel FL consists of the phase duration $t_{3.1}$ and the waiting period $t_{3.2}$. The **handling time t_{30}** depends on the operator and is required to connect the bleeding bottle respectively to open and close the bleeding screw.

The with x marked **timing values** depend on the size of the brake system, the adjusted bleeding pressure respectively the conditions at the vehicle (e.g. accessibility of the bleeding screws, etc). These times must be determined by trials executed at the customer assembly plant. The handling time must be determined together with the operators. These trials are in responsibility of the customer and have to be executed by them self.

All with y marked **pressure and timing values** are to be inferred from the chapter **Process data (pressure and time values)** and **Diagnostic data (timing values)**.

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3.4 Process step 4: Bleeding of the wheel circuit front right (FR)

Bleeding of the wheel circuit front right (FR)	
Operator:	**Diagnostic unit / ECU:**
☞ **Connect the bleeding bottle and open the bleeding screw at the wheel FR.** ☞ **Actuate the brake pedal alternately with a normal speed (actuated for the time t_{BP1}; released for the time t_{BP2}). This step could also automated by a pneumatic or electrical pedal actuation tool.** ☞ **Start the diagnostics (valve activations, etc.) via the tester and wait till the end message is displayed.**	**Specified valve activations plus the defined waiting time for the wheel FR are executed.**
☞ **Close the bleeding screw at the wheel FR and disconnect the bleeding bottle.** ☞ **Stop the brake pedal actuation.**	

Process step:	Parameter:		Physical values:
4	**Bleeding time t_4** (consist of the diagnostic times):	phase duration $t_{4.1\ ABS}$ respectively. $t_{4.1\ ESP}$	**y s**
		waiting period $t_{3.2}$	**x s**
	System pressure p_1		**y bar**
	Time t_{BP1}		**y s**
	Time t_{BP2}		**y s**
	Handling time t_{40}		**x s**

Notes:

The **bleeding screw FR** is opened all other bleeding screws are closed. The brake fluid is pushed out by the pressure of the bleeding device through the ABS/ESP hydraulic unit, the brake pipe, wheel brake cylinder respectively calliper and the bleeding screw. To support this process **the brake pedal** has been **alternate actuated** (actuated for the **time t_{BP1}**; released for the **time t_{BP2}**). This brake pedal stroking creates the **system pressure p_1**, which supports the bleeding process significantly.

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Analogously the **specified valve and pump motor activations** for the **wheel FR** are executed for the **phase duration $t_{4.1_ABS}$** respectively**. $t_{4.1_ESP}$** to bleed the secondary circuit of the ABS/ESP hydraulic unit.

After the specified valve activations are finished, the wheel FR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. The therefore needed additional **waiting period $t_{4.2}$** depends on the vehicle respectively on the design of the base brake system. This time must be specified by the customer. The total **bleeding time t_4** for the wheel FR consists of the phase duration $t_{4.1_ABS}$ respectively $t_{4.1_ESP}$ and the waiting period $t_{4.2}$. The **handling time t_{40}** depends on the operator and is required to connect the bleeding bottle respectively to open and close the bleeding screw.

The with x marked **timing values** depend on the size of the brake system, the adjusted bleeding pressure respectively the conditions at the vehicle (e.g. accessibility of the bleeding screws, etc). These times must be determined by trials executed at the customer assembly plant. The handling time must be determined together with the operators. These trials are in responsibility of the customer and have to be executed by them self.

All with y marked **pressure and timing values** are to be inferred from the chapter **Process data (pressure and time values)** and **Diagnostic data (timing values)**.

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3.5 Process step 5: Bleeding of the wheel circuit rear right (RR)

Bleeding of the wheel circuit rear right (RR)	
Operator:	**Diagnostic unit / ECU:**
☞ **Connect the bleeding bottle and open the bleeding screw at the wheel RR.** ☞ **Actuate the brake pedal alternately with a normal speed (actuated for the time t_{BP1}; released for the time t_{BP2}). This step could also automated by a pneumatic or electrical pedal actuation tool.** ☞ **Start the diagnostics (valve activations, etc.) via the tester and wait till the end message is displayed.**	**Specified valve activations plus the defined waiting time for the wheel RR are executed.**
☞ **Close the bleeding screw at the wheel RR and disconnect the bleeding bottle.** ☞ **Stop the brake pedal actuation.**	

Process step:	Parameter:		Physical values:
5	**Bleeding time t_5** (consist of the diagnostic times):	phase duration $t_{5.1\ ABS}$ respectively. $t_{5.1\ ESP}$	**y s**
		waiting period $t_{5.2}$	**x s**
	System pressure p_1		**y bar**
	Time t_{BP1}		**y s**
	Time t_{BP2}		**y s**
	Handling time t_{50}		**x s**

Notes:

The **bleeding screw RR** is opened all other bleeding screws are closed. The brake fluid is pushed out by the pressure of the bleeding device through the ABS/ESP hydraulic unit, the brake pipe, wheel brake cylinder respectively calliper and the bleeding screw. To support this process **the brake pedal** has been **alternate actuated** (actuated for the **time t_{BP1}**; released for the **time t_{BP2}**). This brake pedal stroking creates the **system pressure p_1**, which supports the bleeding process significantly.

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Analogously the **specified valve and pump motor activations** for the **wheel RR** are executed for the **phase duration $t_{5.1_ABS}$** respectively**. $t_{5.1_ESP}$** to bleed the secondary circuit of the ABS/ESP hydraulic unit.

After the specified valve activations are finished, the wheel RR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. The therefore needed additional **waiting period $t_{4.2}$** depends on the vehicle respectively on the design of the base brake system. This time must be specified by the customer. The total **bleeding time t_5** for the wheel RR consists of the phase duration $t_{5.1_ABS}$ respectively $t_{5.1_ESP}$ and the waiting period $t_{5.2}$. The **handling time t_{50}** depends on the operator and is required to connect the bleeding bottle respectively to open and close the bleeding screw.

The with x marked **timing values** depend on the size of the brake system, the adjusted bleeding pressure respectively the conditions at the vehicle (e.g. accessibility of the bleeding screws, etc). These times must be determined by trials executed at the customer assembly plant. The handling time must be determined together with the operators. These trials are in responsibility of the customer and have to be executed by them self.

All with y marked **pressure and timing values** are to be inferred from the chapter **Process data (pressure and time values)** and **Diagnostic data (timing values)**.

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3.6 Process step 6: Bleeding of the wheel circuit rear left (RL) - optional

Bleeding of the wheel circuit rear left (RL)	
Operator:	**Diagnostic unit / ECU:**
☞ **Connect the bleeding bottle.** ☞ **Repeat following process more than one:** • **Actuate the brake pedal.** • **Open the bleeding screw at the wheel RL.** • **Close the bleeding screw at the wheel RL.** • **Release the brake pedal.**	---
☞ **Disconnect the bleeding bottle.**	---

Process step:	Parameter:	Physical values:
6	System pressure p_1	y bar
	Bleeding time t_6	x s
	Handling time t_{60}	x s

Notes:

To push out the residual air which is might remained back in the wheel brake circuit after the standard bleeding following procedure should be carry out:
At first the bleeding screw at the wheel RL should be closed before the brake pedal is actuated. Then the brake pedal has to be powerful actuated to create a high system pressure. As next the bleeding screw has to be opened. Caused by the high system pressure the residual air is compressed and released within a short time after the bleeding screw was opened. The released air and the brake fluid are pushed out by the pressure of the bleeding device through the ABS/ESP hydraulic unit, the brake pipe, wheel brake cylinder respectively calliper and the bleeding screw. The therefore required **bleeding time t_6** has to be determined by trials at the customer plant. These trials are in responsibility of the customer and have to be executed by them self.

The **handling time t_{60}** depends on the operator and is required to connect the bleeding bottle respectively to open and close the bleeding screw.

The with x marked **timing values** depend on the size of the brake system, the adjusted bleeding pressure respectively the conditions at the vehicle (e.g. accessibility of the bleeding screws, etc). These times must be determined by trials executed at the customer assembly plant. The handling time must be determined together with the operators. These trials are in responsibility of the customer and have to be executed by them self.

The with y marked **pressure value** is to be inferred from the chapter **Process data (pressure and time values)** and **Diagnostic data (timing values)**.

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3.7 Process step 7: Bleeding of the wheel circuit front left (FL) - optional

Bleeding of the wheel circuit front left (FL)	
Operator:	**Diagnostic unit / ECU:**
☞ **Connect the bleeding bottle.** ☞ **Repeat following process more than one:** • **Actuate the brake pedal.** • **Open the bleeding screw at the wheel FL.** • **Close the bleeding screw at the wheel FL.** • **Release the brake pedal.**	---
☞ **Disconnect the bleeding bottle.**	---

Process step:	Parameter:	Physical values:
7	System pressure p_1	y bar
	Bleeding time t_7	x s
	Handling time t_{70}	x s

Notes:

To push out the residual air which is might remained back in the wheel brake circuit after the standard bleeding following procedure should be carry out:
At first the bleeding screw at the wheel FL should be closed before the brake pedal is actuated. Then the brake pedal has to be powerful actuated to create a high system pressure. As next the bleeding screw has to be opened. Caused by the high system pressure the residual air is compressed and released within a short time after the bleeding screw was opened. The released air and the brake fluid are pushed out by the pressure of the bleeding device through the ABS/ESP hydraulic unit, the brake pipe, wheel brake cylinder respectively calliper and the bleeding screw. The therefore required **bleeding time t_7** has to be determined by trials at the customer plant. These trials are in responsibility of the customer and have to be executed by them self.

The **handling time t_{70}** depends on the operator and is required to connect the bleeding bottle respectively to open and close the bleeding screw.

The with x marked **timing values** depend on the size of the brake system, the adjusted bleeding pressure respectively the conditions at the vehicle (e.g. accessibility of the bleeding screws, etc). These times must be determined by trials executed at the customer assembly plant. The handling time must be determined together with the operators. These trials are in responsibility of the customer and have to be executed by them self.

The with y marked **pressure value** is to be inferred from the chapter **Process data (pressure and time values)** and **Diagnostic data (timing values)**.

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3.8 Process step 8: Handling 2

Handling 2		
Operator:		
☞ **Release the vehicle** ☞ **Switch off the bleeding device and disconnect it from the brake fluid reservoir** ☞ **Switch off the ignition** ☞ **Disconnect the electrical connections (diagnostic connector)** ☞ **Correct the brake fluid level respectively fill it till the MAX sign is reached** ☞ **Mount the wheels of the vehicle**		
Process step:	**Parameter:**	**Physical values:**
8	**Handling time t_8**	**x s**

Notes:

The with x marked **handling time t_8** depends on the installation position of the brake fluid reservoir, the position of the diagnosis interface OBD2 in the vehicle interior and the conditions at the vehicle. The handling time t_1 must be determined together with the operators by trials. These trials are in responsibility of the customer and have to be executed by them self.

At first the operator has to switch off the ignition before the diagnostic connector is disconnected.

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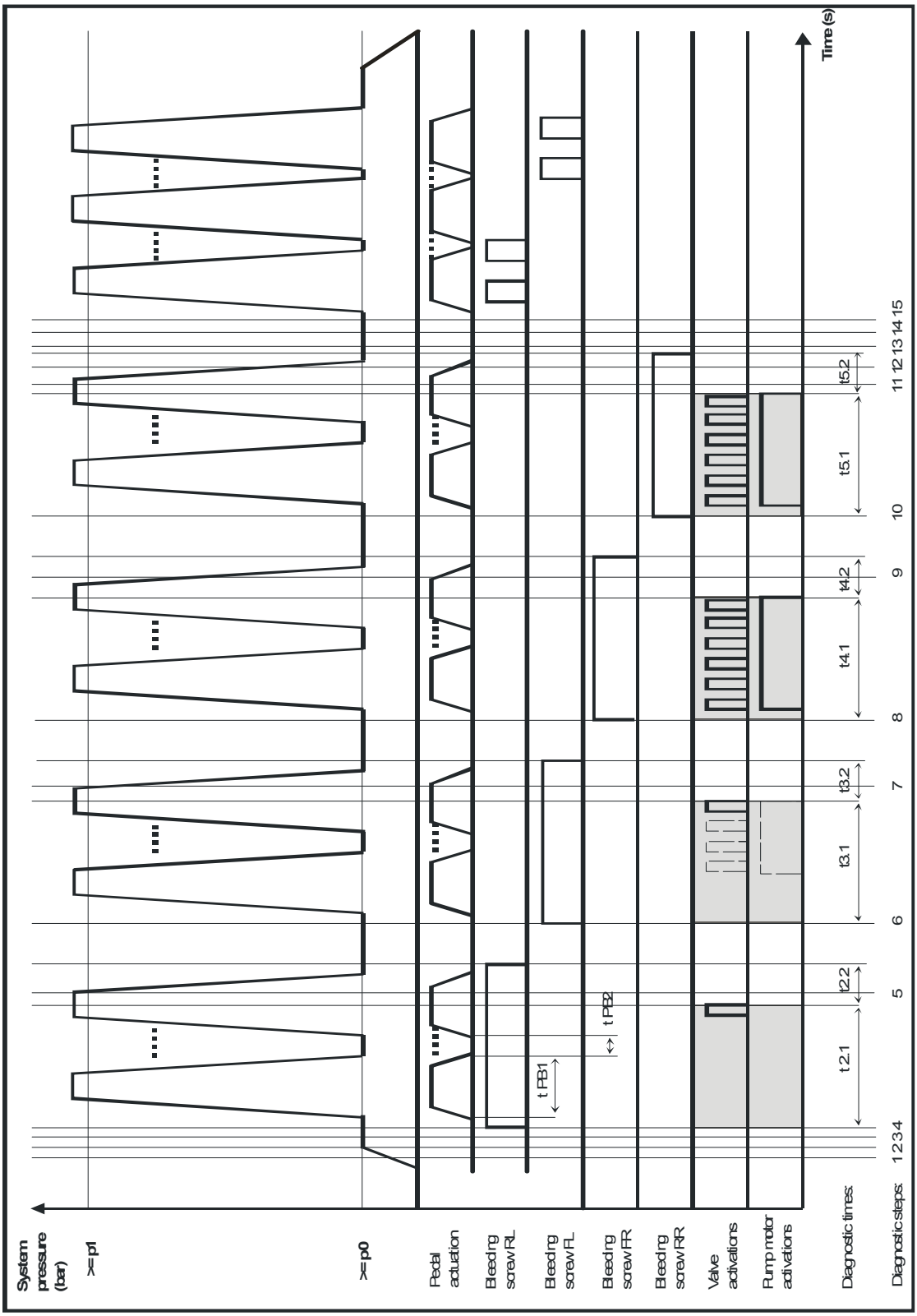
4 Description of all required diagnostic steps

Diagram of the sequence of all required diagnostic functions, valve and pump motor activations for generation 9:



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Notes:

The bleeding screw of the respective wheel has to stay opened as long as the **wheel associated valve and pump motor activations** are finished.
After that the respective wheel has to bleed a additional time as long as the brake fluid is free of air bubbles. The therefore required **waiting periods $t_{2.2}$, $t_{3.2}$, $t_{4.2}$ and $t_{5.2}$** depend on the size of the base brake system. These times have to be determined by trials with serial parts. These trials are in responsibility of the customer and have to be executed by them self.

The details to the valve and pump motor activations for the wheel

rear left (RL) = valve activations for the wheel RL = phase duration $t_{2.1}$ + waiting period $t_{2.2}$,

front left (FL) = valve activations (respectively pump motor activation for a 3 channel ABS) for the wheel FL = phase duration $t_{3.1}$ + waiting period $t_{3.2}$,

front right (FR) = valve and pump motor activations for the wheel FR = phase duration $t_{4.1_ABS}$ respectively $t_{4.1_ESP}$ + waiting period $t_{4.2}$,

rear right (RR) = valve and pump motor activations for the wheel RR = phase duration t $t_{5.1_ABS}$ respectively $t_{5.1_ESP}$ + waiting period $t_{5.2}$

can be taken from the chapter

Diagnosis step 4: Start of the valve activations for the wheel RL

Diagnosis step 6: Start of the valve activations (respectively pump motor activation at a 3 channel ABS) for the wheel FL

Diagnosis step 8: Start of the valve and pump motor activations for the wheel FR

Diagnosis step 10: Start of the valve and pump motor activations for the wheel RR.

The wheel-associated valve and pump motor activations were defined by Bosch to ensure the best possible bleeding result with regard to the ABS / ESP hydraulic modulator. Deviations to the given sequence can lead to a deterioration of the entire bleeding result.

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4.1 Diagnosis step 1: Start of communication

Start of communication	
Diagnostic unit	**ECU**
Start of communication	**Change into the customer specific diagnostic mode**
More information about the chapter "start of communication" is to be inferred from the customer specific documentations.	

Notes:

The communication between the tester and ECU can be take place immediately after switching on of the power supply or ignition.

In case of a not successful execution of the function „start of communication", it should be repeated minimum 3 times before the process is interrupted.

More details to the start of communication are to be inferred from the customer specific documentations.

Start of communication successful ⇒ **Continue**
Start of communication not successful ⇒ **Abort, print out fault message**

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4.2 Diagnosis step 2: ECU identification

ECU identification	
Diagnostic unit	**ECU**
ECU identification	**Output of the requested ECU data.**
More information about the chapter "ECU identification" is to be inferred from the customer specific documentations.	

Notes:

The to be evaluated ECU identification data must be confirmed by the customer and its assembly plants.

The ECU identification data must be compared with the in the tester software stored desired ECU data. This is required to select the right valve activation sequence. The ECU identification data might be changed in case of an ABS/ESP software update respectively modification, therefore it must be easily possible with editor menu to update these data.

More details to the chapter "ECU identification" are to be inferred from the customer specific documentations.

Known system ⇒ **Continue**
Unknown system ⇒ **Abort, print out fault message**

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4.3 Diagnosis step 3: Evaluation of the fault code memory

Evaluation of the fault code memory	
Diagnostic unit	**ECU**
Evaluation of all relevant fault code entries before the process is started. **(all for the repair bleed process relevant fault codes can be taken from the table in the chapter Diagnosis step 12: Evaluation of the fault code memory.**	**Output of all stored fault codes**
More information about the chapter "Evaluation of the fault code memory" is to be inferred from the customer specific documentations.	

Notes:

First of all the fault code memory should be checked for relevant fault codes, because different faults cannot be detected by evaluation of the valve and pump motor status information. More details to the evaluation of the fault code memory can be taken from the chapter Diagnosis step 12: Evaluation of the fault code memory or the customer specific documentations.

None relevant fault code stored ⇒ **Continue**
Relevant fault code stored ⇒ **Abort, print out fault message**

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4.4 Diagnosis step 4: Start of the valve activations for the wheel RL

Start of the valve activations	
Diagnostic unit	**ECU**
Start of the valve activations for the wheel RL	**Activation of the valves**
More information about the chapter "Start of the valve activations for the wheel RL" is to be inferred from the customer specific documentations.	

Notes:

The required valve activations for the wheel **RL** depend on the brake split configuration respectively the system (ABS/ESP). Details to the possible brake split configurations can be taken from the chapter Table of valve assignment depending on the brake circuit configuration.
.

After starting of the repair bleeding process with a **4-channel ABS / ESP system** the valve activations for the wheel RL will be started with activation of the **valve $AV_{P1.1}$** after a defined **waiting time t_{AV0}** for the **valve activation time t_{AV1}**. This is required to press brake fluid in a part of the secondary circuit of the ABS / ESP hydraulic modulator.
After the specified valve activations are finished, RL must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{2.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_2** (= diagnosis time) for the wheel RL consists of the **phase duration $t_{2.1}$** and **waiting time $t_{2.2}$**.

At a 3-channel ABS System no valve activations are executed **at the wheel RL, therefore the whole bleeding time t_2** (= diagnosis time) only consists of the **waiting time $t_{2.2}$**.

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Diagram for the wheel RL for a **4-channel** ABS and ESP system:

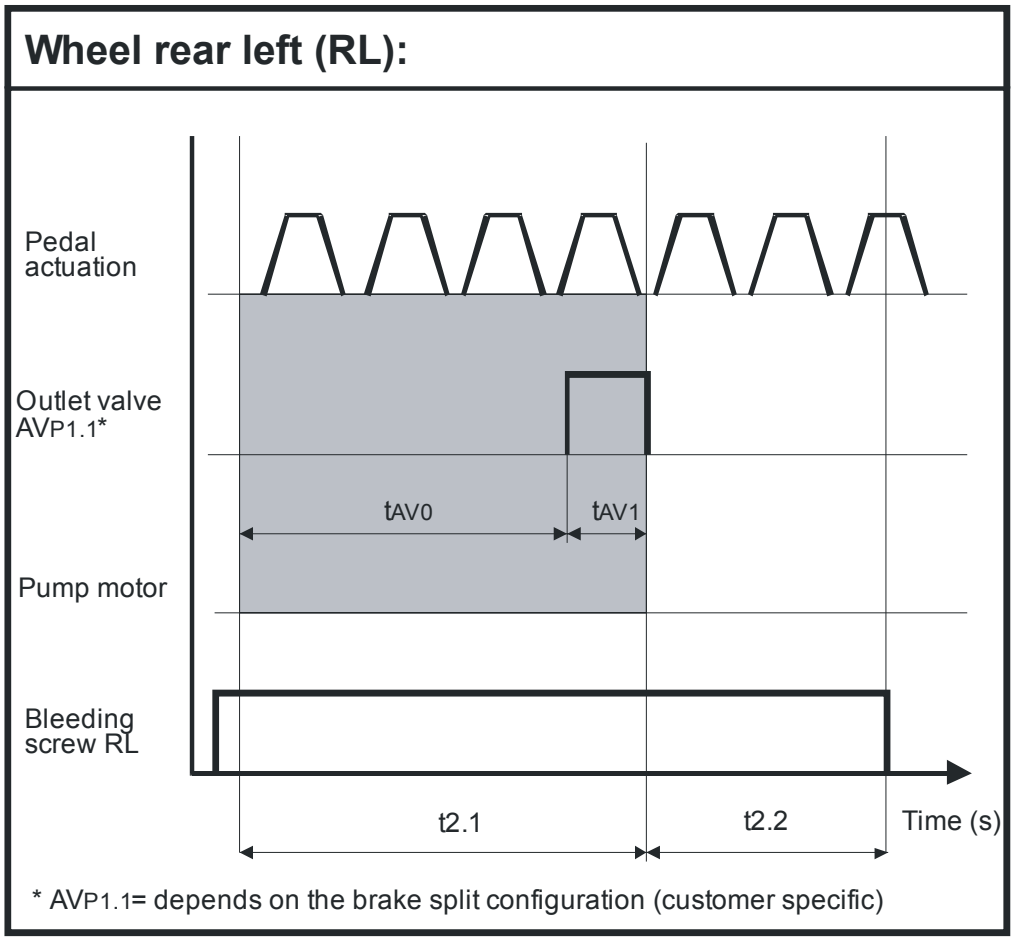

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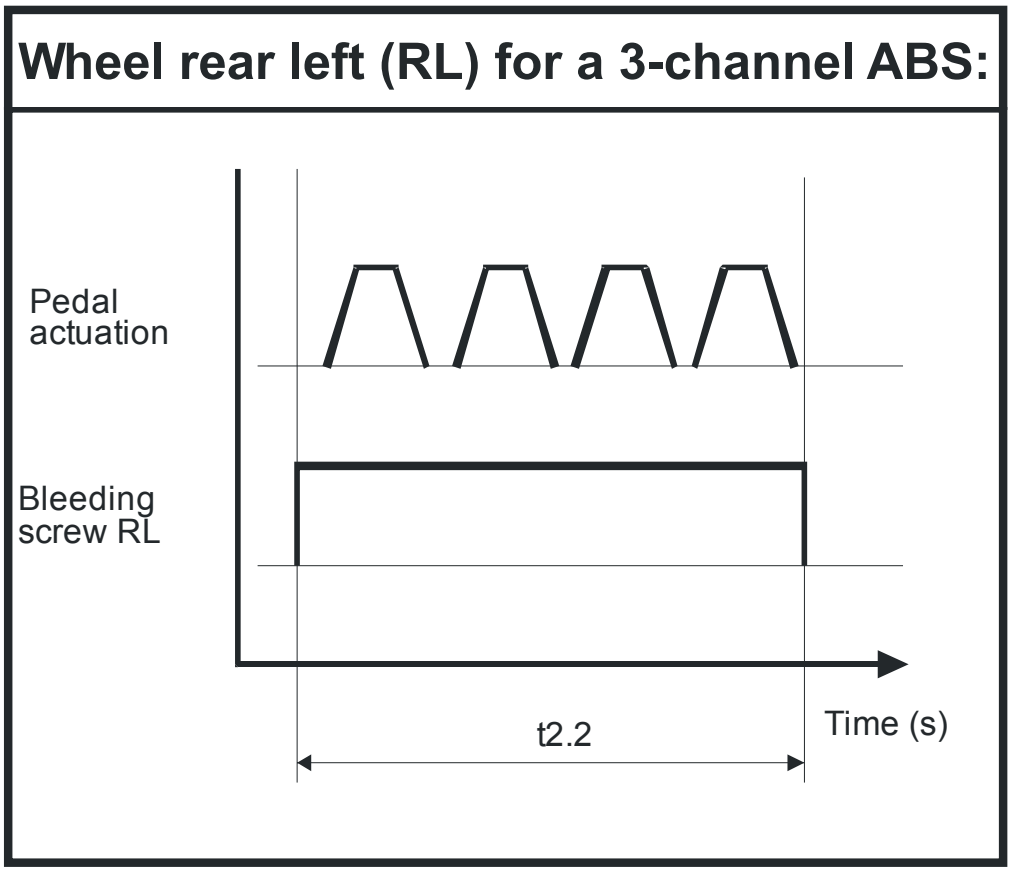
Diagram for the wheel RL for a **3-channel** ABS system:



Diagnostic parameter:

Parameter:	Physical values:
Valve activation time t_{AV0}	**y s**
Valve activation time t_{AV1}	**y s**
Phase duration $t_{2.1}$	**y s**
Waiting time $t_{2.2}$	**x s**

Notes:

The with x marked time value depends on the size of the base brake system and pressure p_0. This time has to be determined by trails executed at the customer plant. The responsibility for these trials has the customer.

All with y marked **time values** are to be inferred from the chapter **Diagnostic data (timing values)**.

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4.5 Diagnosis step 5: Evaluation of the status of the valve activation

Evaluation of the status of the valve activation	
Diagnostic unit	**ECU**
Evaluation of the status of the valve activation	**Output of the status of the valve activation**
More information about the chapter "Evaluation of the status of the valve activation" is to be inferred from the customer specific documentations.	

Notes:

To verify if the valve activation was successful executed and finished, the status of the valve activations have to be evaluated. The required diagnostic function is customer specific.

More details to the chapter "Evaluation of the status of the valve activation" can be taken from the customer specific documentations.

Following ECU responses (examples) are possible:

Valve activation is still running
Valve activation is successful finished
Valve activation interrupted
Valve activation is unsuccessful finished (fault detected)

Important note to the interrogation of the system status (valve relay):

With generation 9 the valve relay is always switched off in the normal state. In dependence of the selected diagnostic mode or first with starting of valve activation the valve relay is turned on. Therefore an interrogation of the valve relay to judge the system status is only limited helpful. However this evaluation can be applied to ensure a correct activation nevertheless in the individual or customer specific case, but this important remark must always be considered.

Status of the valve activation correct ⇒ **Continue**

Status of the valve activation during the repair bleeding process incorrect ⇒ **Abort, print out fault message**

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4.6 Diagnosis step 6: Start of the valve activations (respectively pump motor activation at a 3 channel ABS) for the wheel FL

Start of the valve activations	
Diagnostic unit	**ECU**
Start of the valve activations for the wheel FL	**Activation of the valves**
More information about the chapter "Start of the valve activations (respectively pump motor activation at a 3 channel ABS) for the wheel FL" is to be inferred from the customer specific documentations.	

Notes:

The required valve activations for the wheel **FL** depend on the brake split configuration respectively the system (ABS/ESP). Details to the possible brake split configurations can be taken from the chapter Table of valve assignment depending on the brake circuit configuration.
.

After starting of the repair bleeding process with a **4-channel ABS / ESP system** the valve activations for the wheel FL will be started with activation of the **valve $AV_{P2.1}$** after a defined **waiting time t_{AV0}** for the **valve activation time t_{AV1}**. This is required to press brake fluid in a part of the secondary circuit of the ABS / ESP hydraulic modulator.
After the specified valve activations are finished, FL must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{3.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_3** (= diagnosis time) for the wheel FL consists of the **phase duration $t_{3.1}$** and **waiting time $t_{3.2}$**.

After starting of the repair bleeding process with a **3-channel ABS system** the valve activations for the wheel FL will be started with activation of the valves **$EV_{P2.1}$**, **$EV_{P2.2}$** and **$AV_{P2.1}$** and the **pump motor** after a defined **waiting time t_{AV0}**. The pump motor and the valves $EV_{P2.1}$ and $EV_{P2.2}$ are activated for the **pump motor activation time $t_{PM1.1}$** (= valve activation time). The activation of the pump motor is required to supply the pump element at the rear axle circuit with brake fluid. To the same time the valve $AV_{P2.1}$ is alternate activated on and off for the **valve activation time t_{AVon}** and **valve activation time** t_{AVoff}. After that the valve **$AV_{P2.2}$** is activated for the **valve activation time t_{AV1}**.
After the specified valve activations are finished, FL must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{3.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_3** (= diagnosis time) for the wheel FL consists of the **phase duration** $t_{3.1_3K_ABS}$ and **waiting time $t_{3.2}$**.

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Diagram for the wheel FL for a **4-channel** ABS and ESP system:

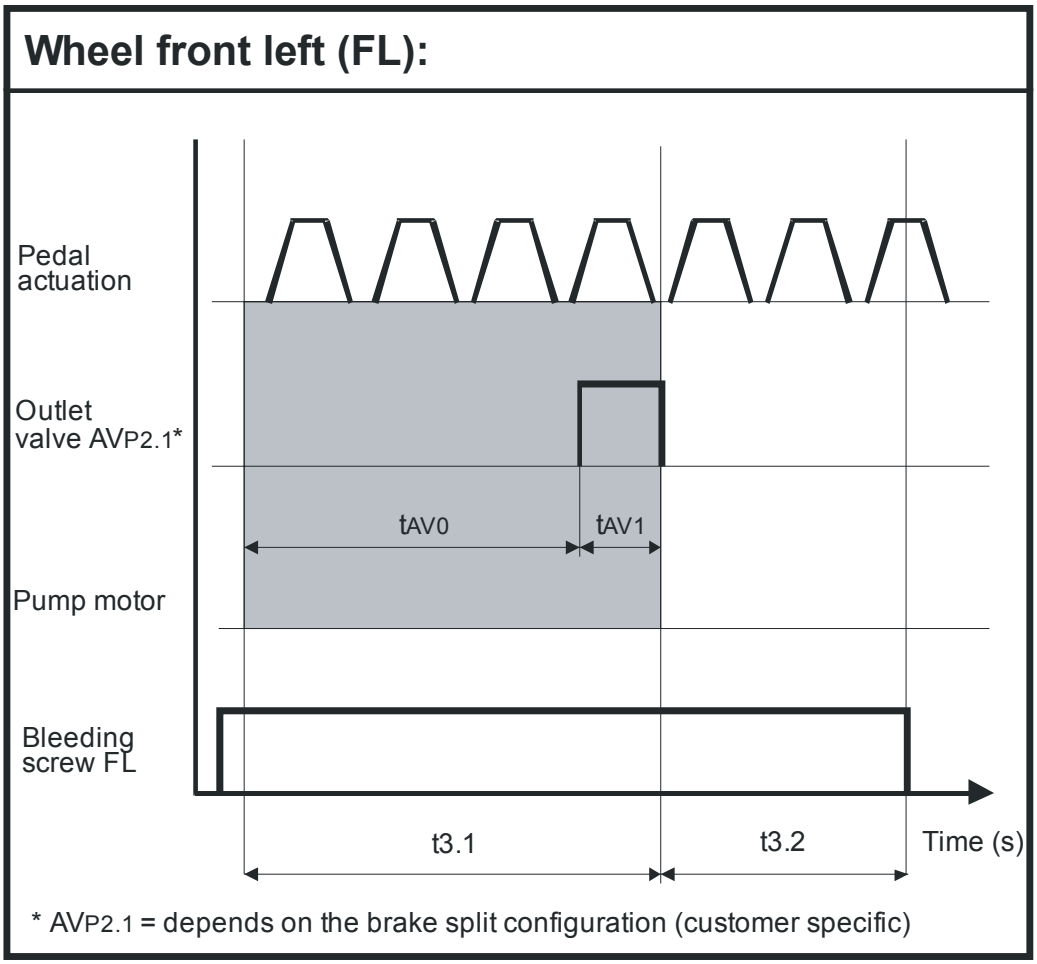

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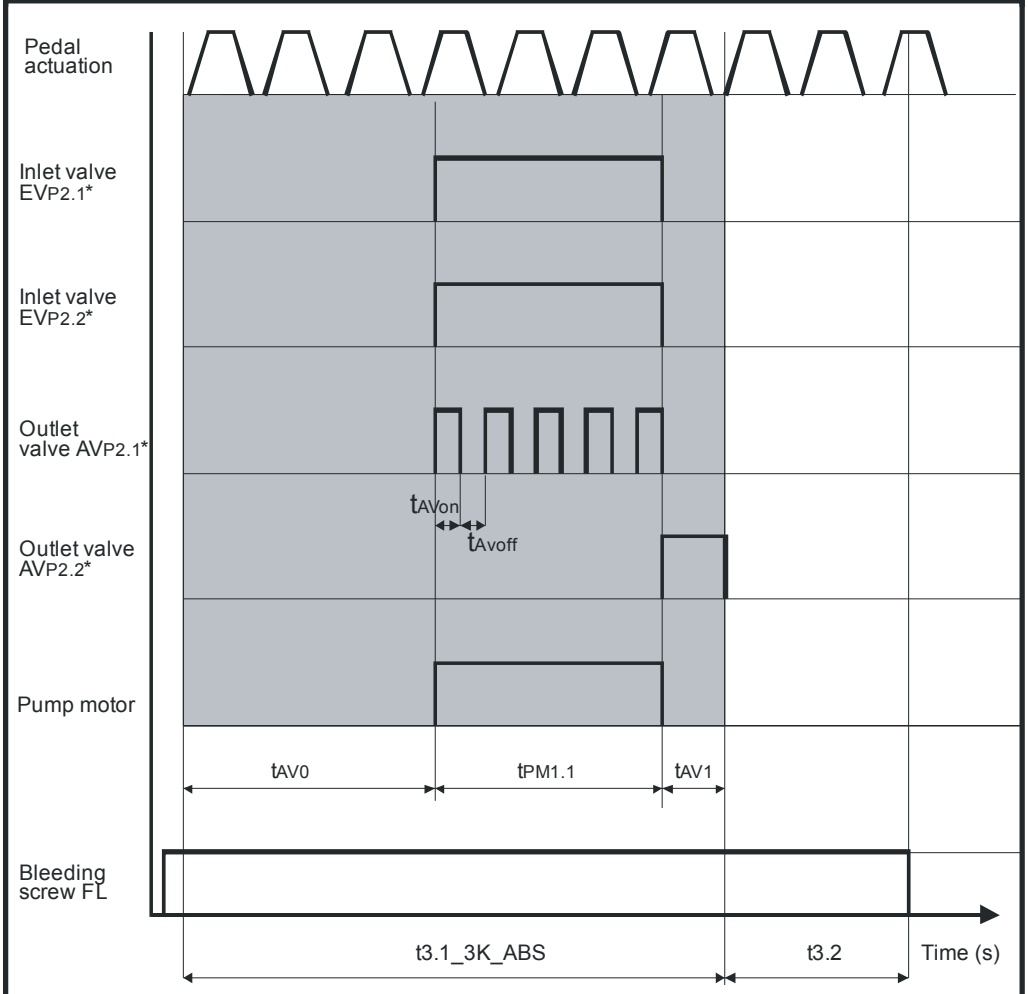
Diagram for the wheel FL for a **3-channel** ABS system:

Wheel (FL) for a 3-channel ABS:



All with * marked valves depend on the brake split configuration (customer specific)

Diagnostic parameter:

Parameter:	Physical values:
Valve activation time t_{AV0}	**y s**
Valve activation time t_{AV1}	**y s**
Valve activation time t_{AVon}	**y s**
Valve activation time t_{AVoff}	**y s**
Pump motor activation time $t_{PM1.1}$	**y s**
Phase duration $t_{3.1}$	**y s**
Phase duration $t_{3.1_3K_ABS}$	**y s**
Waiting time $t_{3.2}$	**x s**

Notes:

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The with x marked time value depends on the size of the base brake system and pressure p_0. This time has to be determined by trails executed at the customer plant. The responsibility for these trials has the customer.
All with y marked **time values** are to be inferred from the chapter **Diagnostic data (timing values)**.

4.7 Diagnosis step 7: Evaluation of the status of the valve activation

Details can be taken from the chapter 4.5.

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4.8 Diagnosis step 8: Start of the valve and pump motor activations for the wheel FR

Start of the valve activations	
Diagnostic unit	**ECU**
Start of the valve activations or the wheel FR	**Activation of the valves**
More information about the chapter "Start of the valve and pump motor activations for the wheel FR" is to be inferred from the customer specific documentations.	

Notes:

The required valve activations for the wheel **FR** depend on the brake split configuration respectively the system (ABS/ESP). Details to the possible brake split configurations can be taken from the chapter Table of valve assignment depending on the brake circuit configuration.

Description for a 3-channel ABS system:

After starting of the repair bleeding process the valve activations for the wheel FR will be started with activation of the valves **$EV_{P3.1}$**, **$EV_{P3.2}$** and **$AV_{P3.1}$** after a defined **waiting time t_{Delay1}** for the **valve activation time t_{AV2}**.
After a further waiting time (= valve activation time t_{AV2} – pump motor activation time $t_{PM1.1}$) the pump motor is activated for the **pump motor activation time $t_{PM1.1}$**.The activation of the pump motor is required to supply the pump element at the front axle circuit with brake fluid.
Immediately after all above described activations are finished the bleeding of the respective secondary circuit is started with execution of the valve activations which are included in a loop. This loop is repeated for a defined number (number ≤ n1). Following valve activations are included in this loop:
After a defined **waiting time t_{Delay1}** the pump motor is activated for the **pump motor activation time $t_{PM1.2}$**. Directly afterwards the **waiting times** t_{Delay1} and **t_{Delay2}** are expired the valve **$AV_{P3.2}$** is switched on for the **valve activation time t_{AV1}**. After the valve activation time t_{AV1} is expired the valve **$AV_{P3.1}$** is activated with the same time (valve activation time t_{AV1}).
After the selected number of loops respectively all valve activations are finished and a further **waiting time t_{Delay1}** is expired, the pump motor is activated for the **pump motor activation time $t_{PM1.4}$** to discharge the accumulator chambers.
Afterwards the specified valve activations are finished, FR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{4.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_4** (= diagnosis time) for the wheel FR consists of the **phase duration $t_{4.1_ABS}$** and **waiting time $t_{4.2}$**.

Description for a 4-channel ABS system:

After starting of the repair bleeding process the valve activations for the wheel FR will be started with activation of the valves **$EV_{P3.1}$**, **$EV_{P3.2}$**, **$AV_{P3.1}$** and **$AV_{P3.2}$** after a defined **waiting time t_{Delay1}** for the **valve activation time t_{AV2}**.
After a further waiting time (= valve activation time t_{AV2} – pump motor activation time $t_{PM1.1}$) the pump motor is activated for the **pump motor activation time $t_{PM1.1}$**.The activation of the pump motor is required to supply the pump elements with brake fluid.

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Immediately after all above described activations are finished the bleeding of the respective secondary circuit is started with execution of the valve activations which are included in a loop. This loop is repeated for a defined number (number ≤ n1). Following valve activations are included in this loop:
After a defined **waiting time t_{Delay1}** the pump motor is activated for the **pump motor activation time $t_{PM1.2}$**. Directly afterwards the **waiting times** t_{Delay1} and **t_{Delay2}** are expired the valve **$AV_{P3.3}$** is switched on for the **valve activation time t_{AV1}**. After the valve activation time t_{AV1} is expired the valve **$AV_{P3.1}$** is activated with the same time (valve activation time t_{AV1}).
After the selected number of loops respectively all valve activations are finished and a further **waiting time t_{Delay1}** is expired, the pump motor is activated for the **pump motor activation time $t_{PM1.4}$** to discharge the accumulator chambers.
Afterwards the specified valve activations are finished, FR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{4.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_4** (= diagnosis time) for the wheel FR consists of the **phase duration $t_{4.1_ABS}$** and **waiting time $t_{4.2}$**.

<u>Description for an ESP system:</u>

After starting of the repair bleeding process the valve activations for the wheel FR will be started with activation of the valves **$EV_{P3.1}$** and **$EV_{P3.2}$** after a defined **waiting time t_{Delay1}** for the **valve activation time t_{AV3}**. In parallel the valves **$AV_{P3.1}$** and **$AV_{P3.2}$** are activated **t_{Delay1}** for the **valve activation time t_{AV1}**. Directly after the valve activation time t_{AV1} is expired the high pressure switching valves **$HSV_{P3.1}$** and **$HSV_{P3.2}$** are alternate activated with on/ off **(t_{HSVon} / t_{HSVoff})** for the **valve activation time $t_{HSV1.1}$**. **$AV_{P3.3}$** is switched on for the **valve activation time t_{AV1}**. In addition the pump motor is activated for the **pump motor activation time $t_{PM1.1}$**.The activation of the pump motor is required to supply the pump elements with brake fluid.
Immediately after all above described activations are finished the bleeding of the respective secondary circuit is started with execution of the valve activations which are included in a loop. This loop is repeated for a defined number (number ≤ n1). Following valve activations are included in this loop:
After a defined **waiting time t_{Delay1}** the pump motor is activated for the **pump motor activation time $t_{PM1.2}$**. Directly afterwards the **waiting times** t_{Delay1} and **t_{Delay2}** are expired the valve **$AV_{P3.3}$** is switched on for the **valve activation time t_{AV1}**. After the valve activation time t_{AV1} is expired the valve **$AV_{P3.1}$** is activated with the same time (valve activation time t_{AV1}).
After the selected number of loops respectively all valve activations are finished and a further **waiting time t_{Delay1}** is expired the high pressure switching valve **$HSV_{P3.1}$** is alternate activated with on/ off **(t_{HSVon} / t_{HSVoff})** for the **valve activation time $t_{HSV1.2}$**. In parallel the pump motor is activated for the **pump motor activation time $t_{PM1.3}$** to bleed the respective priming circuit. In fact of that the pump motor activation time $t_{PM1.3}$ is longer than the valve activation time $t_{HSV1.2}$ all valves are closed at the end, therefore it is guaranteed that the accumulators will be discharged.
Afterwards the specified valve activations are finished, FR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{4.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_4** (= diagnosis time) for the wheel FR consists of the **phase duration $t_{4.1_ESP}$** and **waiting time $t_{4.2}$**.

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Diagram for the wheel FR for a **3-channel** ABS system:

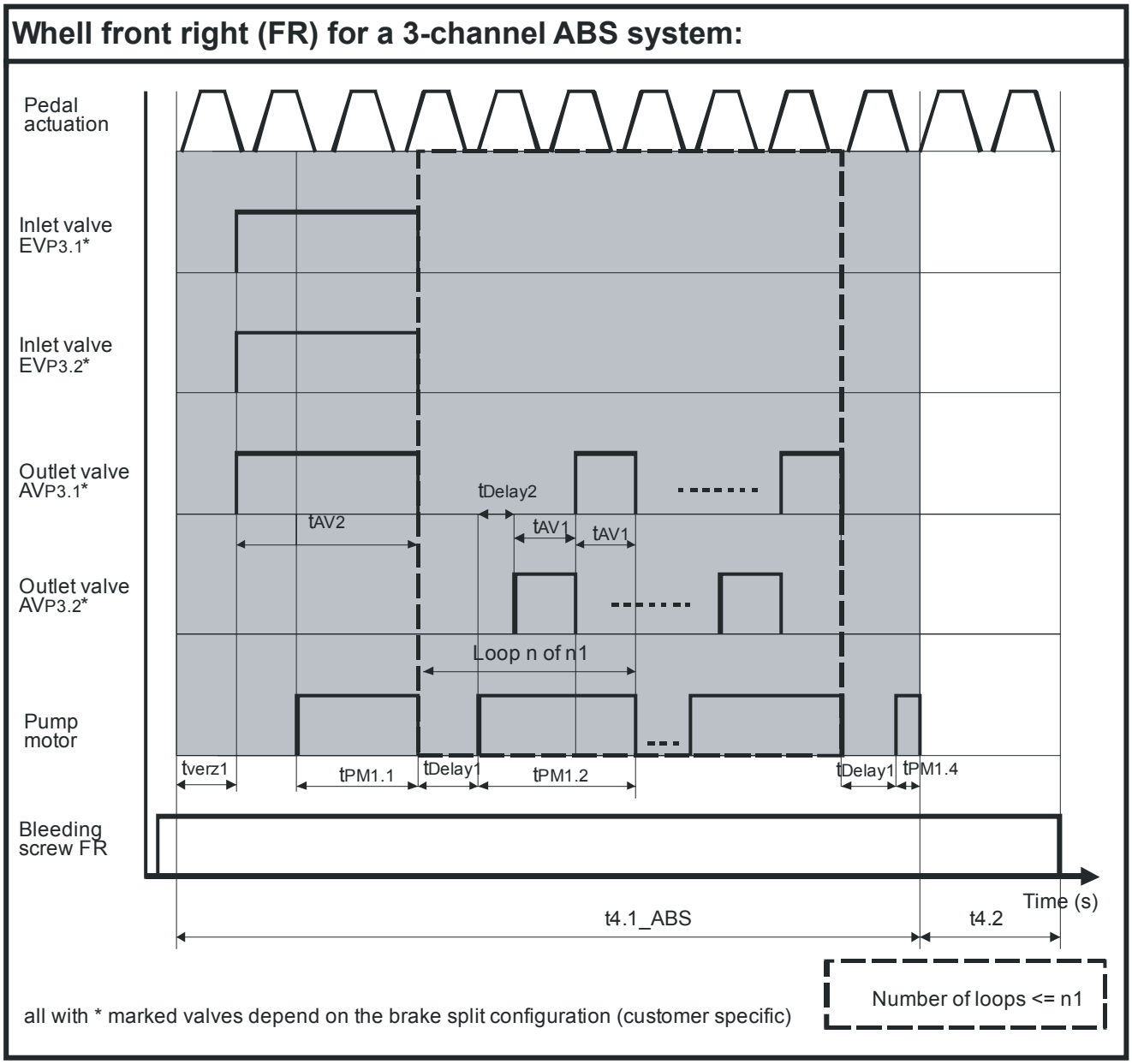

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Diagram for the wheel FR for a **4-channel** ABS system:

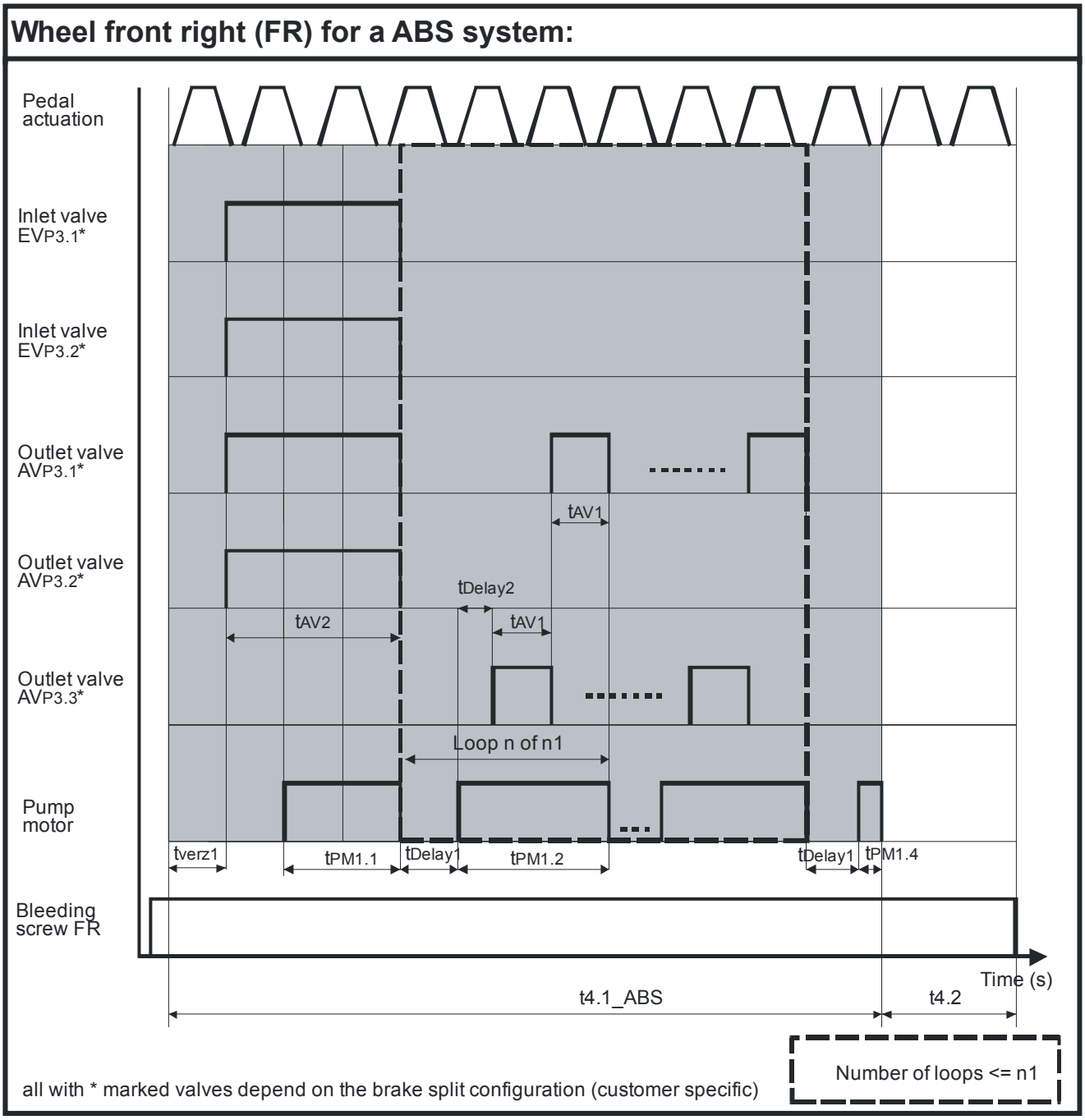

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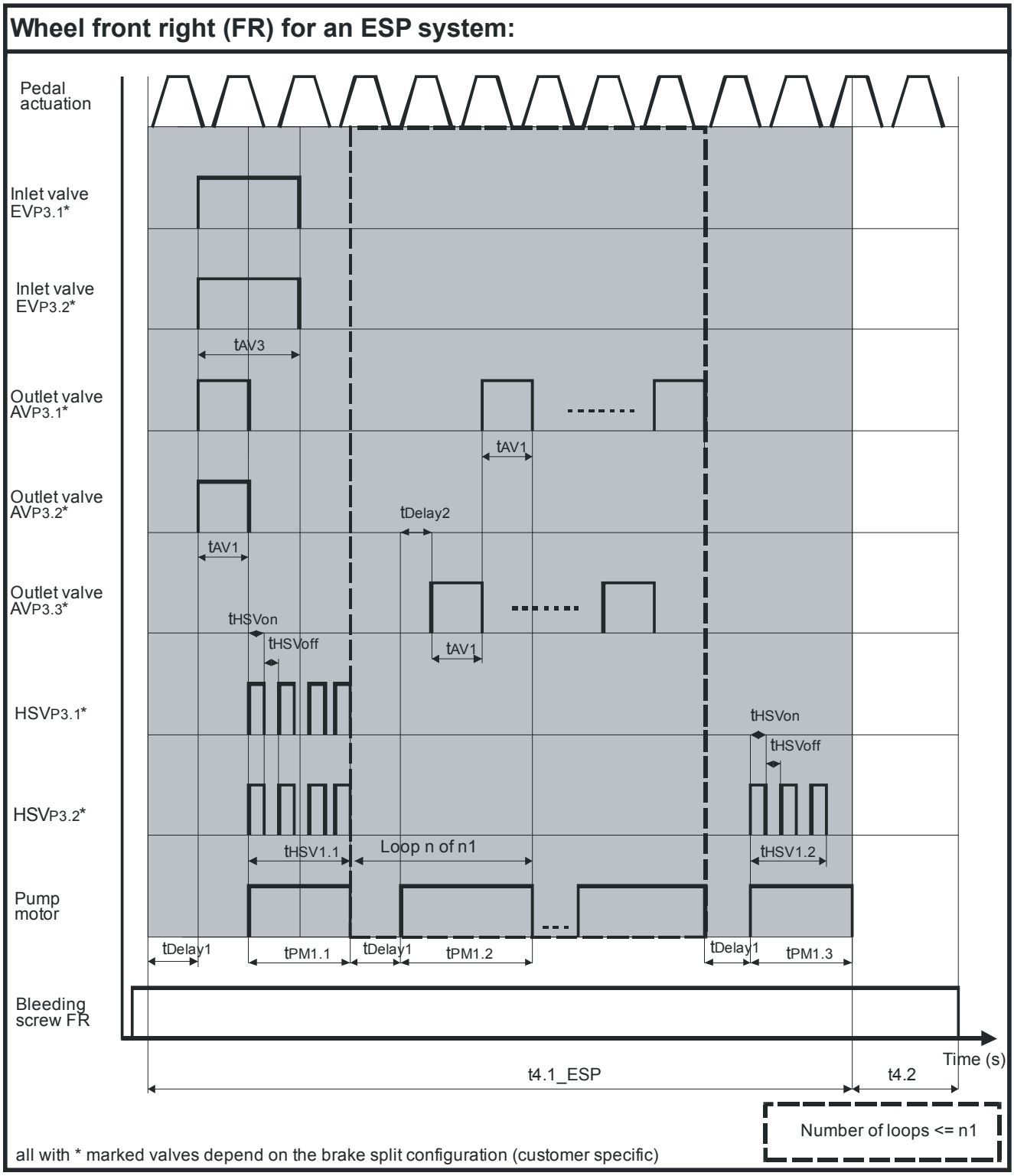
Diagram for the wheel FR for an **ESP** system:



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Diagnostic parameter:

Parameter:	Physical values:
Valve activation time t_{AV1}	y s
Valve activation time t_{AV2}	y s
Valve activation time t_{AV3}	y s
Valve activation time $t_{HSV1.1}$	y s
Valve activation time $t_{HSV1.2}$	y s
Valve activation time t_{HSVon}	y s
Valve activation time t_{HSVoff}	y s
Waiting time t_{Delay1}	y s
Waiting time t_{Delay2}	y s
Pump motor activation time $t_{PM1.1}$	y s
Pump motor activation time $t_{PM1.2}$	y s
Pump motor activation time $t_{PM1.3}$	y s
Pump motor activation time $t_{PM1.4}$	y s
Number of loops n_1	y
Max. Phase duration $t_{4.1_ABS}$	y s
Max. Phase duration $t_{4.1_ESP}$	y s
Waiting time $t_{4.2}$	x s

Notes:

The with x marked time value depends on the size of the base brake system and pressure p_0. This time has to be determined by trails executed at the customer plant. The responsibility for these trials has the customer.
All with y marked **time values** are to be inferred from the chapter **Diagnostic data (timing values)**.

4.9 Diagnosis step 9: Evaluation of the status of the valve activation

Details can be taken from the chapter 4.5.

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4.10 Diagnosis step 10: Start of the valve and pump motor activations for the wheel RR

Start of the valve activations	
Diagnostic unit	**ECU**
Start of the valve activations or the wheel RR	**Activation of the valves**
More information about the chapter "Start of the valve and pump motor activations for the wheel RR" is to be inferred from the customer specific documentations.	

Notes:

Description for a 3-channel ABS system:

After starting of the repair bleeding process for the wheel RR the bleeding of the respective secondary circuit is started with execution of the valve activations which are included in a loop. This loop is repeated for a defined number (number ≤ n1). Following valve activations are included in this loop:
After a defined **waiting time t_{Delay1}** the pump motor is activated for the **pump motor activation time $t_{PM1.5}$**. Directly afterwards the **waiting times** t_{Delay1} and **t_{Delay2}** are expired the valve **$AV_{P4.1}$** is switched on for the **valve activation time t_{AV1}**.
After the selected number of loops respectively all valve activations are finished and a further **waiting time t_{Delay1}** is expired, the pump motor is activated for the **pump motor activation time $t_{PM1.4}$** to discharge the accumulator chambers.
Afterwards the specified valve activations are finished, RR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{5.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_5** (= diagnosis time) for the wheel RR consists of the **phase duration $t_{5.1_ABS}$** and **waiting time $t_{5.2}$**.

Description for a 4-channel ABS system:

After starting of the repair bleeding process for the wheel RR the bleeding of the respective secondary circuit is started with execution of the valve activations which are included in a loop. This loop is repeated for a defined number (number ≤ n1). Following valve activations are included in this loop:
After a defined **waiting time t_{Delay1}** the pump motor is activated for the **pump motor activation time $t_{PM1.2}$**. Directly afterwards the **waiting times** t_{Delay1} and **t_{Delay2}** are expired the valve **$AV_{P4.2}$** is switched on for the **valve activation time t_{AV1}**. After the valve activation time t_{AV1} is expired the valve **$AV_{P4.1}$** is activated with the same time (valve activation time t_{AV1}).
After the selected number of loops respectively all valve activations are finished and a further **waiting time t_{Delay1}** is expired, the pump motor is activated for the **pump motor activation time $t_{PM1.4}$** to discharge the accumulator chambers.
Afterwards the specified valve activations are finished, RR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{5.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_5** (= diagnosis time) for the wheel RR consists of the **phase duration $t_{5.1_ABS}$** and **waiting time $t_{5.2}$**.

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Description for an ESP system:

After starting of the repair bleeding process for the wheel RR the bleeding of the respective secondary circuit is started with execution of the valve activations which are included in a loop. This loop is repeated for a defined number (number ≤ n1). Following valve activations are included in this loop:
After a defined **waiting time t_{Delay1}** the pump motor is activated for the **pump motor activation time $t_{PM1.2}$**. Directly afterwards the **waiting times** t_{Delay1} and **t_{Delay2}** are expired the valve **$AV_{P4.2}$** is switched on for the **valve activation time t_{AV1}**. After the valve activation time t_{AV1} is expired the valve **$AV_{P4.1}$** is activated with the same time (valve activation time t_{AV1}).
After the selected number of loops respectively all valve activations are finished and a further **waiting time t_{Delay1}** is expired the high pressure switching valve **$HSV_{P4.1}$** is alternate activated with on/ off **(t_{HSVon} / t_{HSVoff})** for the **valve activation time $t_{HSV1.2}$**. In parallel the pump motor is activated for the **pump motor activation time $t_{PM1.3}$** to bleed the respective priming circuit. In fact of that the pump motor activation time $t_{PM1.3}$ is longer than the valve activation time $t_{HSV1.2}$ all valves are closed at the end, therefore it is guaranteed that the accumulators will be discharged.
Afterwards the specified valve activations are finished, RR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{4.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_5** (= diagnosis time) for the wheel RR consists of the **phase duration $t_{5.1_ESP}$** and **waiting time $t_{5.2}$**.

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Diagram for the wheel RR for a **3-channel** ABS system:

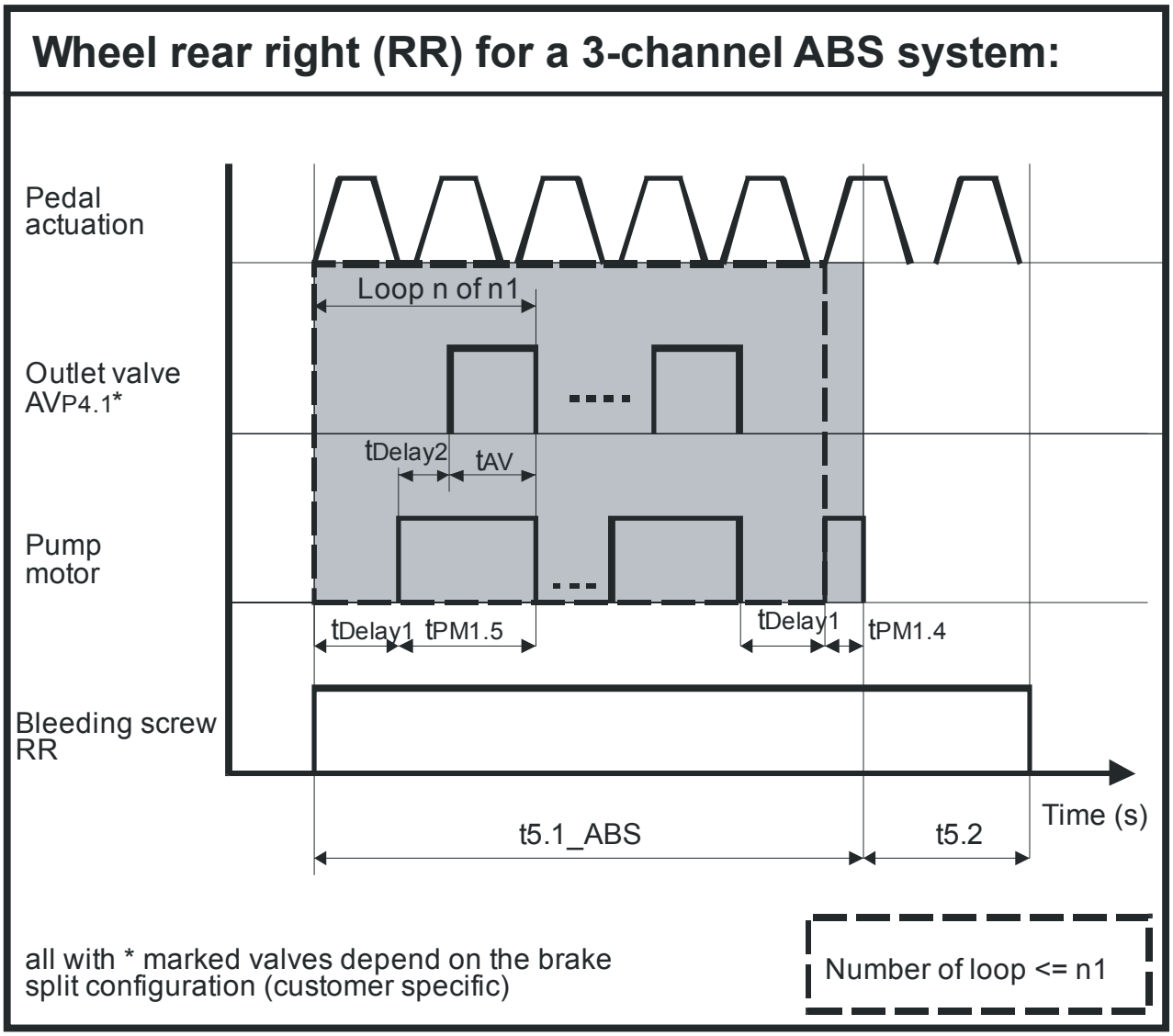

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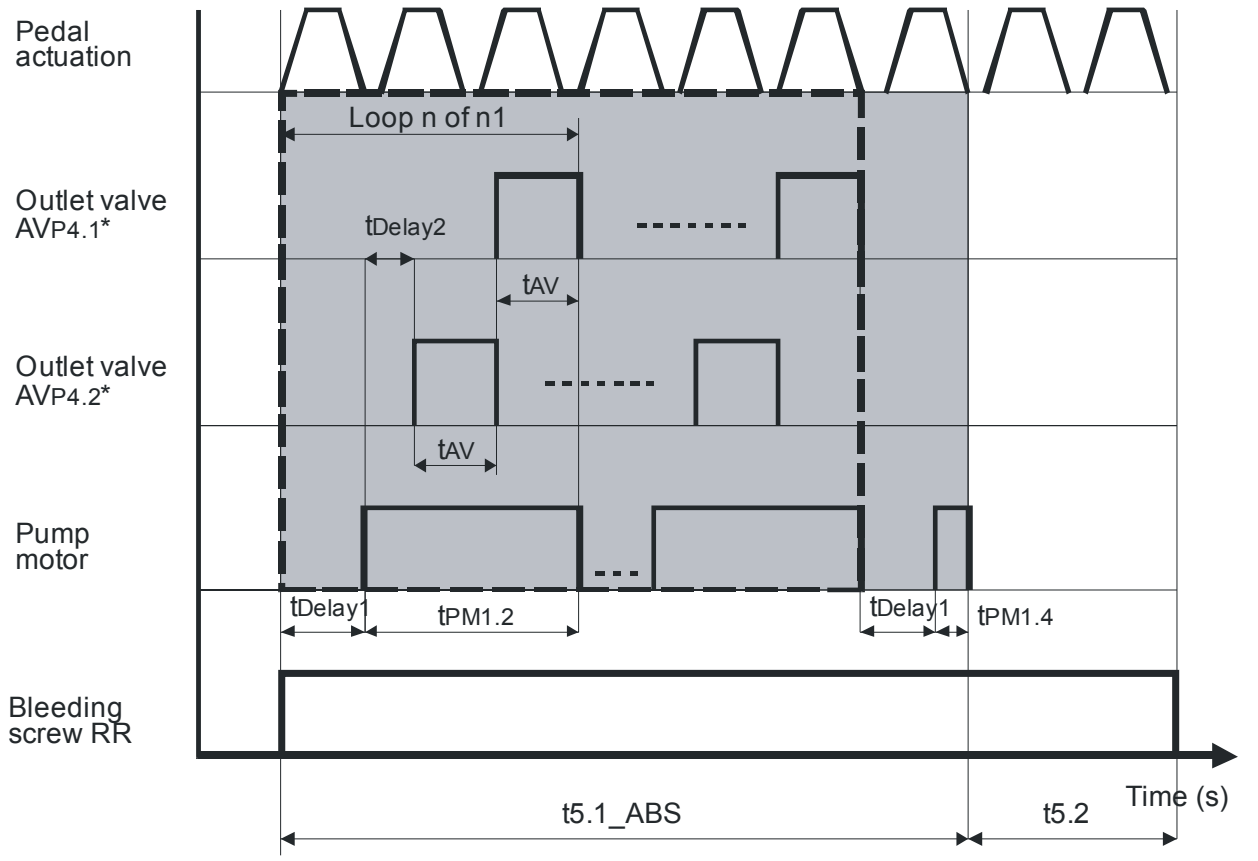
Diagram for the wheel RR for a **4-channel** ABS system:

Wheel rear right (RR) for an ABS system:



all with * marked valves depend on the brake split configuration (customer specific)

Number of loops <= n1

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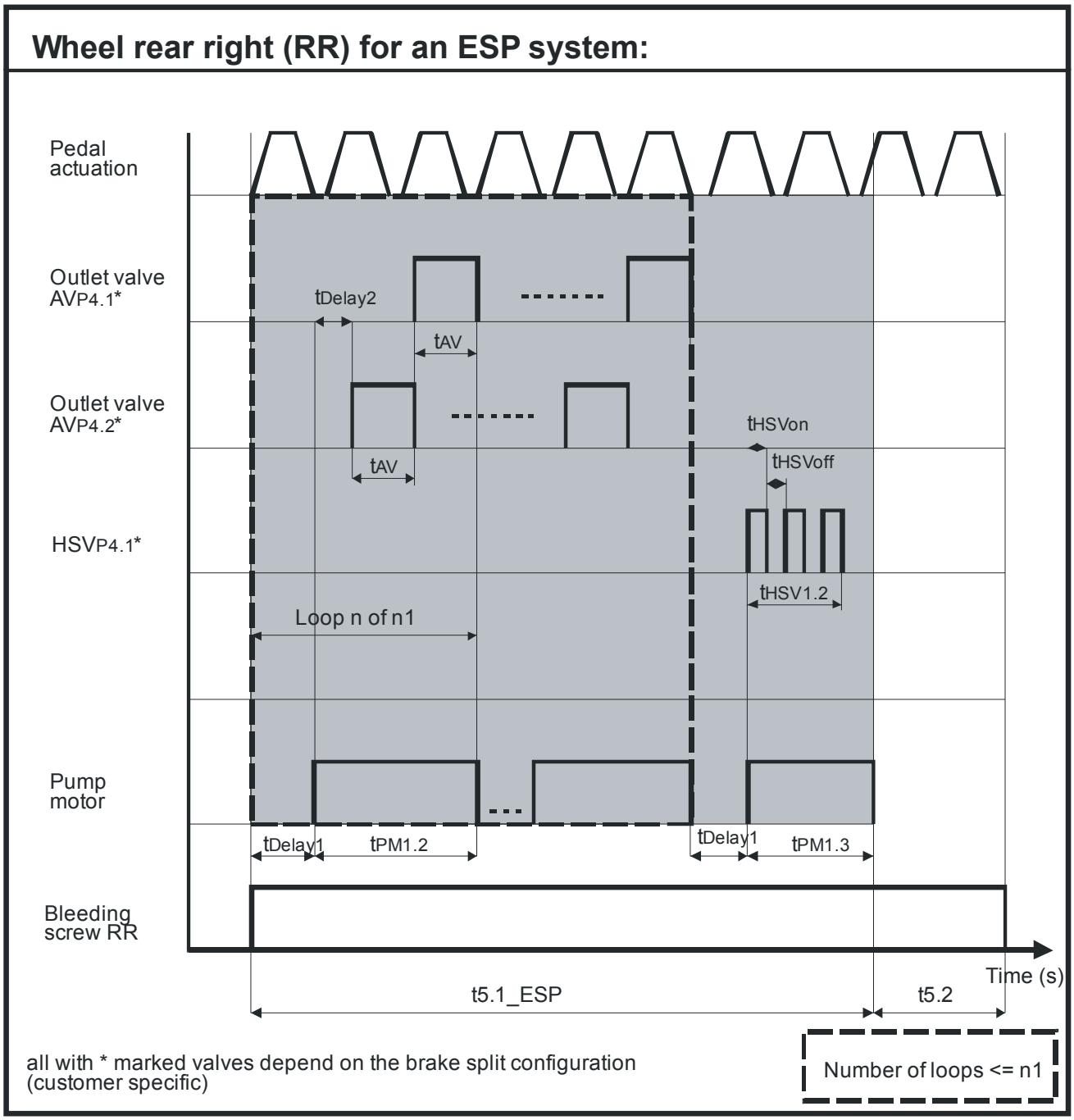
Diagram for the wheel RR for an ESP system:



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Diagnostic parameter:

Parameter:	Physical values:
Valve activation time t_{AV1}	y s
Valve activation time $t_{HSV1.2}$	y s
Valve activation time t_{HSVon}	y s
Valve activation time t_{HSVoff}	y s
Waiting time t_{Delay1}	y s
Waiting time t_{Delay2}	y s
Pump motor activation time $t_{PM1.2}$	y s
Pump motor activation time $t_{PM1.3}$	y s
Pump motor activation time $t_{PM1.4}$	y s
Pump motor activation time $t_{PM1.5}$	y s
Number of loops n_1	y
Max. Phase duration $t_{5.1_ABS}$	y s
Max. Phase duration $t_{5.1_ESP}$	y s
Waiting time $t_{5.2}$	x s

Notes:

The with x marked time value depends on the size of the base brake system and pressure p_0. This time has to be determined by trails executed at the customer plant. The responsibility for these trials has the customer.
All with y marked **time values** are to be inferred from the chapter **Diagnostic data (timing values)**.

4.11 Diagnosis step 11: Evaluation of the status of the valve activation

Details can be taken from the chapter 4.5.

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4.12 Diagnosis step 12: Evaluation of the fault code memory

Evaluation of the fault code memory	
Diagnostic unit	**ECU**
Evaluation of all relevant fault code entries after the process is finished.	**Output of all stored fault codes**
More information about the chapter "Evaluation of the fault code memory" is to be inferred from the customer specific documentations.	

Notes:

To verify if all valves and the pump motor were correct activated in the diagnostic mode, the fault code memory has to be evaluated. Only relevant and actual faults have to be considered.
The ABS/ESP ECU gives out the complete contents of the entire fault code memory. These data consists of the following information:

- Diagnostic trouble code (DTC)
- Status of DTC (actual / historic)

The diagnostic tester has to verify the complete outputted contents of the fault code memory. In case of relevant diagnostic trouble codes are stored, they have to be converted into a failure text and a fault protocol has to print out.
The status information of a stored DTC shows if it is actual or historic.

Only following faults are relevant for a correct executed repair bleeding process all other stored faults (e.g. missing calibration, missing variant code, etc.) can be ignored:

Description of the diagnostic trouble code:	Will be detected by evaluation of:
Outlet valves	**Status of valve activation**
Inlet valves	
High pressure switching valves (only ESP systems)	
Power supply for the valves	
Power supply for the pump motor	**Fault code memory**
Pump motor	

None relevant fault code stored ⇒ **Continue**
Relevant fault code stored ⇒ **Abort, print out fault message**

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4.13 Write process bytes (optional)

Write process bytes	
Diagnostic unit	**ECU**
Write process bytes	**Writes the process results into the EEPROM of the ABS/ESP system**
More information to the chapter "Write process bytes" is to be inferred from the customer specific documentations.	

Notes:

All with **tbd** (to be defined) marked values are customer specific.
With this optional function it is possible to write additional information about the repair bleed process into the EEPROM of the ABS/ESP system.
The ABS/ESP ECU is delivered with predefined value (customer specific). This value is written into the ABS/ESP EEPROM by the Bosch plant. In case of none successful communication with the ABS/ESP ECU at the customer assembly plant, the delivery value is still stored. These conditions must be considered as a not successful respectively faulty repair bleed process.
The offered EEPROM location to store this production information amounts up to two data bytes (customer specific). The definition and to be written process data is in responsibility of the customer. To do the right evaluation all defined values and data must be hand out to the corresponding suppliers (e.g. end of line equipment).

Example of writing two process bytes:

Repair bleed equipment writes the process bytes with detailed information:			
Fault description of the process byte 1:	**Fault description of the process byte 2:**	**PB 1 data to be written (hex):**	**PB 2 data to be written (hex):**
Fault during activation of the valves for the wheel FL **Entire repair bleed process i.O.** **etc.**	**Repair bleed facility 1** **Repair bleed facility 2** **etc.**	**tbd** **tbd** **etc.**	**tbd** **tbd** **etc.**
Repair bleed equipment writes the process bytes without detailed information:			
Fault description of the process byte 1:	**Fault description of the process byte 2:**	**PB 1 data to be written (hex):**	**PB 2 data to be written (hex):**
Process ok **Process nok**	**Default respectively delivery value**	**tbd** **tbd**	**tbd** **tbd**

Write process bytes ok ⇒ **Continue**
Write process bytes nok. ⇒ **Print out fault message, abort**

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4.14 Diagnosis step 14: Clear fault code memory (optional)

Clear fault code memory	
Diagnostic unit	**ECU**
Clearing of the fault code memory	**Fault code memory is erased.**
More information to the chapter "Clear fault code memory (optional)" is to be inferred from the customer specific documentations.	

Notes:

With this step the complete contents of the fault code memory is erased. This step is optional and can also be executed at a later process step.

Clearing of the fault code memory successful ⇒ **Continue**
Clearing of the fault code memory not successful ⇒ **Print out fault message, continue**

4.15 Diagnosis step 15: End of diagnostics

End of diagnostics	
Diagnostic unit	**ECU**
Leaving of the customer specific diagnostic mode.	**Leaves the customer specific diagnostic mode.**
More information to the chapter "End of diagnostics" is to be inferred from the customer specific documentations.	

Notes:

With this step the customer specific diagnostic mode in finished respectively left. This step is required to change back to regular (normal) ABS/ESP control mode. In case of the valve relay is disabled caused by a detected fault a changing of the diagnostic mode is might not be possible (behaviour is customer specific).

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5 Failure handling

Failure handling	
Root cause of failure:	**Failure message, failure handling:**
Bleeding device	**Failure message, mark vehicle with failure label → Repeat the complete repair bleeding at all four wheels**
Diagnostic unit	**Failure message, mark vehicle with failure label → Repeat the complete repair bleeding at all four wheels**

Notes:

In case of any occurred failure the entire repair bleeding process has to be repeated at all four wheels.

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6 Data

6.1 Process data (pressure and time values)

Process step:	Parameter:	Condi-tions:	Physi-cal value:	Unit:
1 to 8	Absolute pressure p_0	≥	3	Bar
		≤	tbd*	
	System pressure p_1	≥	15	Bar
	Time t_{BP1}	~	3	Seconds
	Time t_{BP2}	~	0,5	

Notes:

All with **tbd** (to be defined) marked values are customer specific.

The with x marked maximal allowed absolute pressure p_0 depends on the maximal allowed pressure for the brake fluid reservoir. This value is customer specific.

6.2 Diagnostic data (timing values)

Diagnosis step:	Parameter:	Condi-tions:	Physi-cal value:	Unit:
4	Phase duration $t_{2.1}$	=	25	Seconds
	Waiting period $t_{2.2}$	=	tbd	
6	Phase duration $t_{3.1}$	=	25	
	Phase duration $t_{3.1_3K_ABS}$	=	35	
	Waiting period $t_{3.2}$	=	tbd	
8	Max. Phase duration $t_{4.1_ABS}$	≤	99*	
	Max. Phase duration $t_{4.1_ESP}$	≤	106*	
	Waiting period $t_{4.2}$	=	tbd	
10	Max. Phase duration $t_{5.1_ABS}$	≤	79*	
	Max. Phase duration $t_{5.1_ESP}$	≤	86*	
	Waiting period $t_{5.2}$	=	tbd	

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General valid data for following diagnosis steps:					
4, 6, 8 and 10	Number of loops n_1	*min.*	≥	3	---
		recommended	=	4	
		max.	≤	5	
	Valve activation time t_{AVon}		=	1	Seconds
	Valve activation time t_{AVoff}		=	1	
	Valve activation time t_{AV0}		=	20	
	Valve activation time t_{AV1}		=	5	
	Valve activation time t_{AV2}		=	15	
	Valve activation time t_{AV3}		=	10	
	Valve activation time $t_{HSV1.1}$		=	10	
	Valve activation time $t_{HSV1.2}$		=	7	
	Valve activation time t_{HSVon}		=	1	
	Valve activation time t_{HSVoff}		=	0,5	
	Waiting time t_{Delay1}		=	5	
	Waiting time t_{Delay2}		=	3	
	Pump motor activation time $t_{PM1.1}$		=	10	
	Pump motor activation time $t_{PM1.2}$		=	13	
	Pump motor activation time $t_{PM1.3}$		=	9	
	Pump motor activation time $t_{PM1.4}$		=	2	
	Pump motor activation time $t_{PM1.5}$		=	8	

Notes:

All with **tbd** (to be defined) marked values are customer specific.

The with x marked maximal phase durations $t_{4.1_ABS}$, $t_{4.1_ESP}$, $t_{5.1_ABS}$ and $t_{5.1_ESP}$ are the values for the recommended numbers of loops.

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6.3 Hydraulic data

6.3.1 Table of valve assignment depending on the brake circuit configuration

Diagnosis step:	Valve activation parame-ter:	Diagonal brake split configuration (X):		Front/rear brake split configuration (II)		
		4- channel				3- channel
		ABS	ESP	ABS	ESP	ABS
4	$AV_{P1.1}$	FR		RR		---
6	$AV_{P2.1}$	RR		FR		RA
	$AV_{P2.2}$	---		---		FR
	$EV_{P2.1}$	---		---		FL
	$EV_{P2.2}$	---		---		FR
8	$AV_{P3.1}$	RL		FL		FL
	$AV_{P3.2}$	FL		RL		FR
	$AV_{P3.3}$	FR		FR		---
	$EV_{P3.1}$	FR		FR		RA
	$EV_{P3.2}$	RR		RR		FR
	$HSV_{P3.1}$	---	FL/RR	---	RL/RR	---
	$HSV_{P3.2}$	---	FR/RL	---	FL/FR	---
10	$AV_{P4.1}$	FL		RL		RA
	$AV_{P4.2}$	RR		RR		---
	$HSV_{P4.1}$	---	FL/RR	---	RL/RR	---

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6.3.2 Volume and valve orifice data respectively pump element dry run time

Volume and valve orifice data respectively pump element dry run time:	ABS 9	ESP 9
Volume of the primary circuit per circuit (cm^3):	tbd	tbd
Volume of the secondary circuit per circuit (cm^3):	tbd	tbd
Volume of the priming circuit per circuit (cm^3):	---	tbd
Dead space of the pump elements per circuit (cm^3):	tbd	tbd
Inlet valve orifice diameter RA (mm):	customer specific	
Inlet valve orifice diameter FA (mm):	customer specific	
Outlet valve orifice diameter RA (mm):	customer specific	
Inlet valve orifice diameter FA (mm):	customer specific	
Maximal Pump element dry run time t_{PEdry} (s):	≤ 60	

Notes:

All mentioned volume data are rounded values. The used orifices are project and customer specific.

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6.4 Electrical data

6.4.1 Information to the calculated current consumption of an ABS/ESP system

Calculated current consumption at a 12V voltage supply				
Component:		**Component:**	**Compo-nent:**	**Component:**
ECU:		**≤ 0,5**	**A**	
AV coil (7,3 Ω):		**1,64**	**A**	
EV coil (7,3 Ω):		**1,23**	**A**	**Valve protection measures active! For more details please refer the chapter Valve protection measures at genera-tion 9.**
HSV coil (3,6 Ω):		**3,3/0,97**	**A**	
Max. total valve current consump-tion:	**ABS:**	**6,02**	**A**	**Only valid if the defined valve activa-tion ratio and profile is kept (worst case at an ABS: 2xAV+2xEV, at an ESP: 2xEV+2xHSV to the same time) at each wheel.**
	ESP	**6,40**		
PM start up current I_{on}		**≤ 145**	**A**	**Depending on the used pump motor.**
PM start up current time t_{on}		**≤ 20**	**ms**	
PM constant current I_{on_const}		**≤ 15**	**A**	

6.4.2 Specification of the voltage supply (optional)

Specification voltage supply		
Parameter:	**Physical value:**	**Remarks:**
Range of voltage [V]:	**11,5 ≤ power supply unit ≤ 14**	**Allowed voltage range measured at the ECU. It must be guaranteed that the voltage never drops below 10V (e.g. buffering of the voltage with a condenser required)!**
Current [A]:	**≥ 20**	**To supply the vehicle battery.**
	≥ 145	**To supply the complete vehicle (no battery installed).**

Notes:

The repair bleed equipment respectively its power supply hast to meet the above described require-ments. Independent of the length of the wires between the power supply and the ABS/ESP ECU and the possible caused voltage drop the specified electrical values must be kept.

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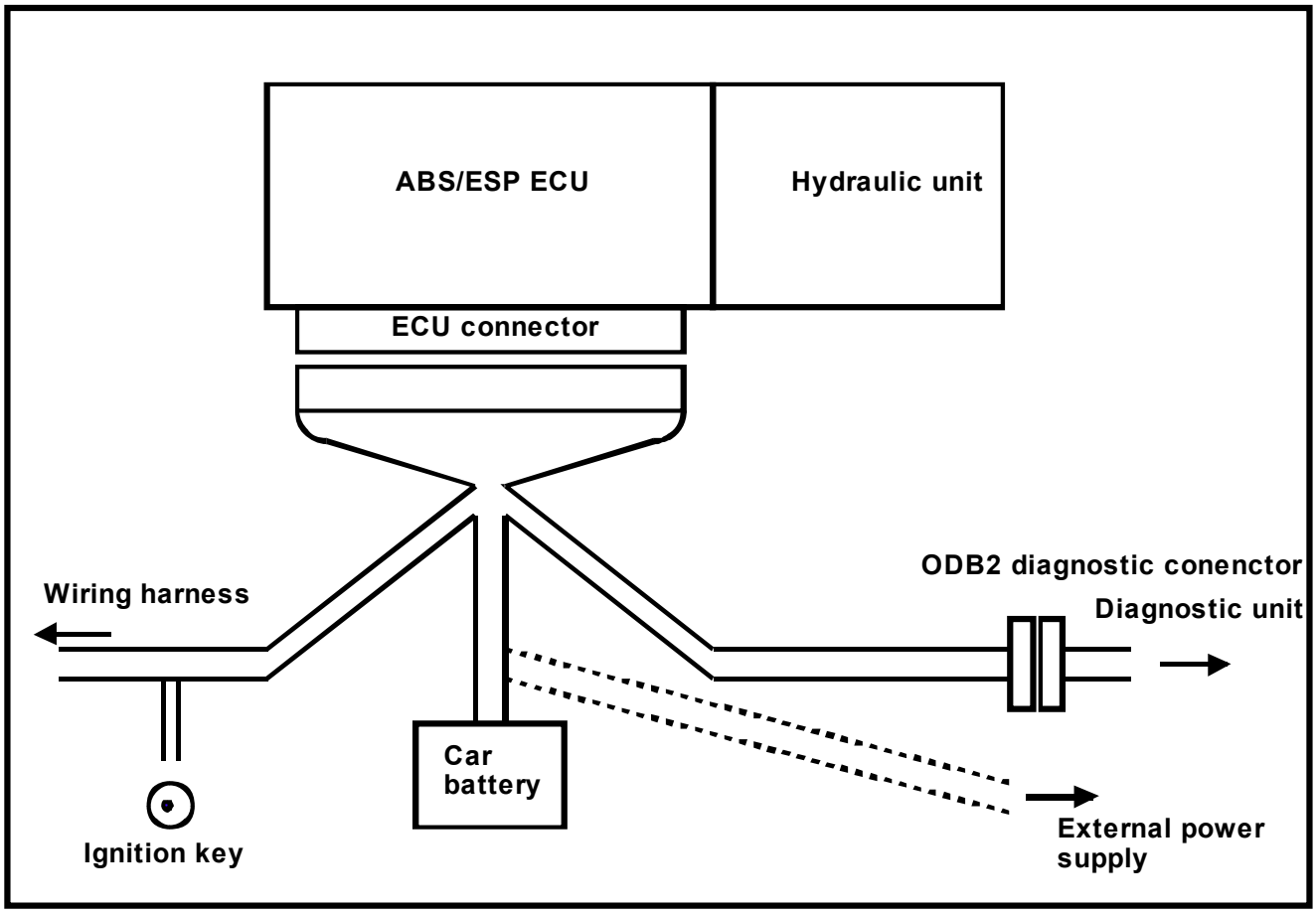
6.4.3 Connection at the diagnostic connector (OBD2) within the vehicle

Handling	
Operator	**Remarks:**
☞ **Adapt diagnostic connector**	**Either in the passenger or engine compartment.**
☞ **Adapt power supply**	**External power supply is required either to supply the car battery or the ABS/ESP ECU itself (e.g. car battery is not installed prior this process step).**
☞ **Switch ignition key to position on**	**To switch on and initialize the ABS/ESP system.**

Notes:

The length of the diagnostic connection wires should be covered and should not exceed 10 meters.



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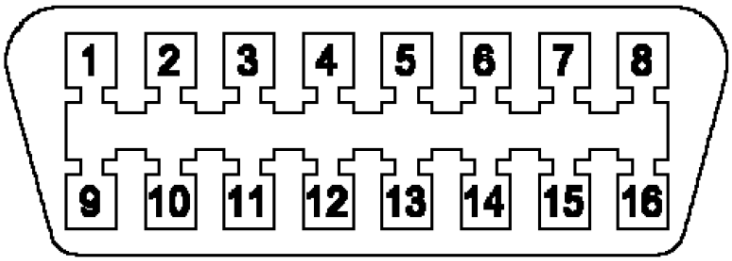
6.4.4 Specification of the diagnostic OBD2 connector within the passenger cell

Specification of the OBD2 diagnostic connector (female):



Description of the Pin:	Pin- No.:
Customer specific	1-3
Vehicle ground	4
Signal ground	5
CAN-High	6
ISO 9141 K-Line	7
Customer specific	8-13
CAN-Low	14
Customer specific	15
+12V power supply	16

Notes:

The used connector has to be an industrial one. It is not recommended to use a simple standard connector, because with a standard connector only limited safe connections are possible. To ensure a safe connection a lot of times at a serial production an industrial connector with golden terminals should is required.

All customer specific information and to be used diagnostic interface have to be taken from the project specific documents.

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7 Valve protection measures at generation 9

	ABS	ESP
Valve:	**Diagnostic mode**	
EV 7,3 Ω	**If UVR > 9V, than lowering of the duty cycle to Ueff = 9V; If valve activation value > 0%; than start the up/down counter and count with a ratio up/down = 2/1; limited to 120s; for each EV valve a separate counter is used**	
AV 7,3 Ω	**If valve activation value > 0% (via customer diagnosis only full time =100% activation possible); than start the up/down counter and count with a ratio up/down = 2/1; limited to 30s; for each AV valve a separate counter is used**	
HSV 3,3 Ω	**---**	**20ms with 100% duty cycle, than lowering of the duty cycle to $U_{effective}$ = 3,5V and cyclic all 200ms for 20ms with 100%**
USV 7,3 Ω	**---**	**Not activated**
USV 4,3 Ω		

Notes:

All valve protection measurements are active in the diagnostic mode. These measurements can not be switched off.

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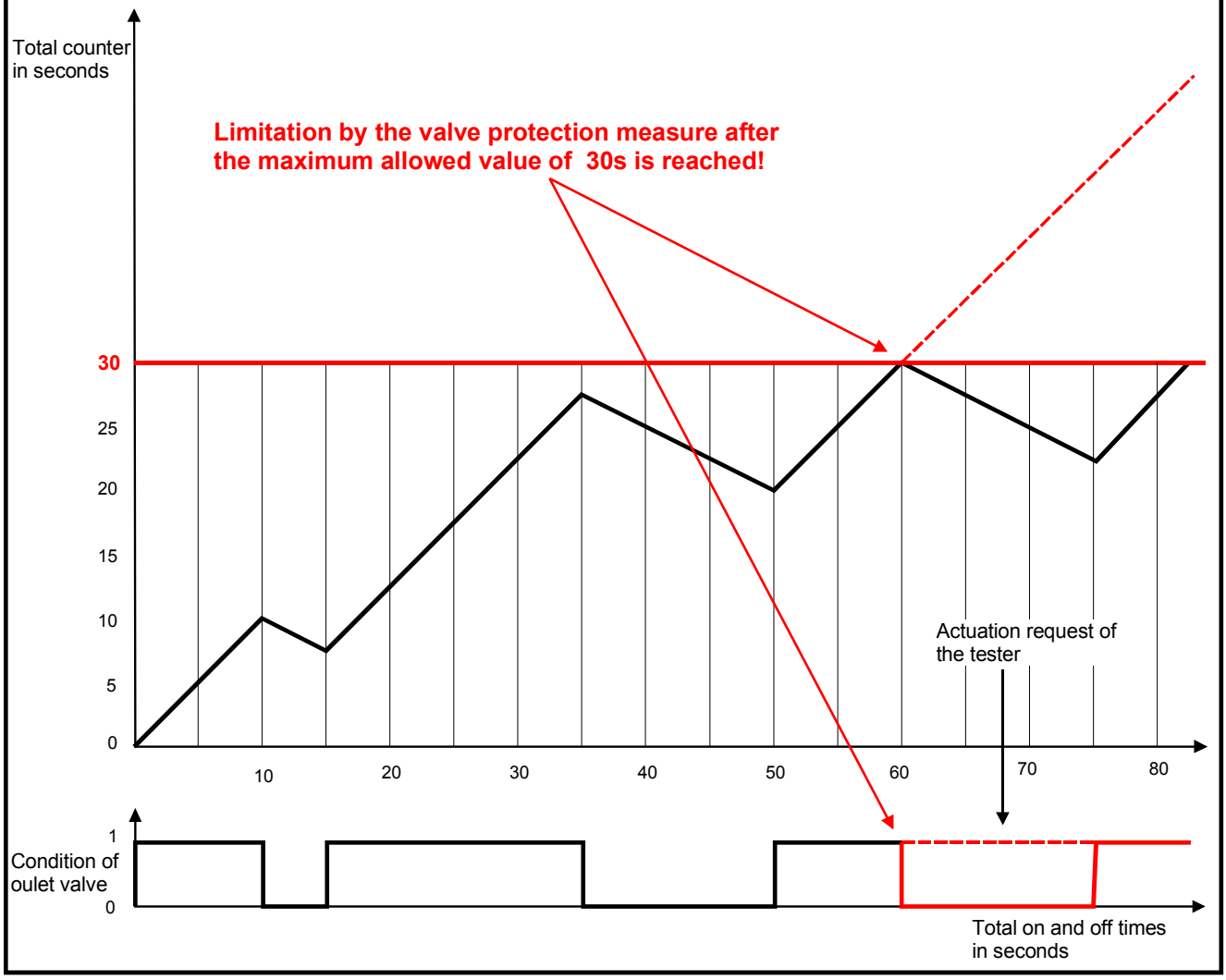
Gen 9: Valve protection measure: Up/down counter with a ratio of = 2/1

Example: Outlet valve with a maximum limitation to 30 seconds



Notes:

A forced limitation by the valve protection measurement is active as long as the valve coils and the ECU are protected against overheating. First after a corresponding cooling time the valves can be switched on again.
To ensure the important coil respectively ECU cooling time after a executed valve actuation via diagnosis in a previous ignition cycle (after a hardware or software reset), the actual coil resistance values are measured and used after each power up to initialise the start up value of the up/down counters.
Caused by different reasons the described valve protection measurements cannot protect the valve coils and the ECU with 100% safeness. That's the reason why the customer is responsible to ensure a correct valve activation sequence!

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8 Equipment information

Diagnostic information

A diagnostic unit is required to activate the valves and the pump motor. This is required to evacuate and fill the secondary circuits of the ABS/ESP system.

To avoid problems with the communication (especially in case of k-line diagnosis) between the diagnostic unit and ABS/ESP ECU, the diagnostic unit should be located as close as possible at the vehicle.

Process bytes

The final definition of the process byte data which should be written at the repair bleed process must be defined by the customer.

Editor menu

The ECU identification data might be changed by software update during a running serial production.

To have the possibility to add these new ECU data at the repair bleed equipment, the editor menu must be designed in a way that a complement of these missing new ECU identification data is easily possible for the operator.

Test protocol

In case of an ok test result, a print out (”test passed”) is made.

In case of a nok test result, a failure protocol must be printed out.

This protocol must include all technical information about the failed step in order to allow a better failure analysis.

An entire test protocol of all process step results must be available on request.

Vehicle conditions

The vehicle battery must be mounted and properly charged in order to allow a sufficient power supply to the ABS/ESP system. A external power supply should only be used to charge the battery.

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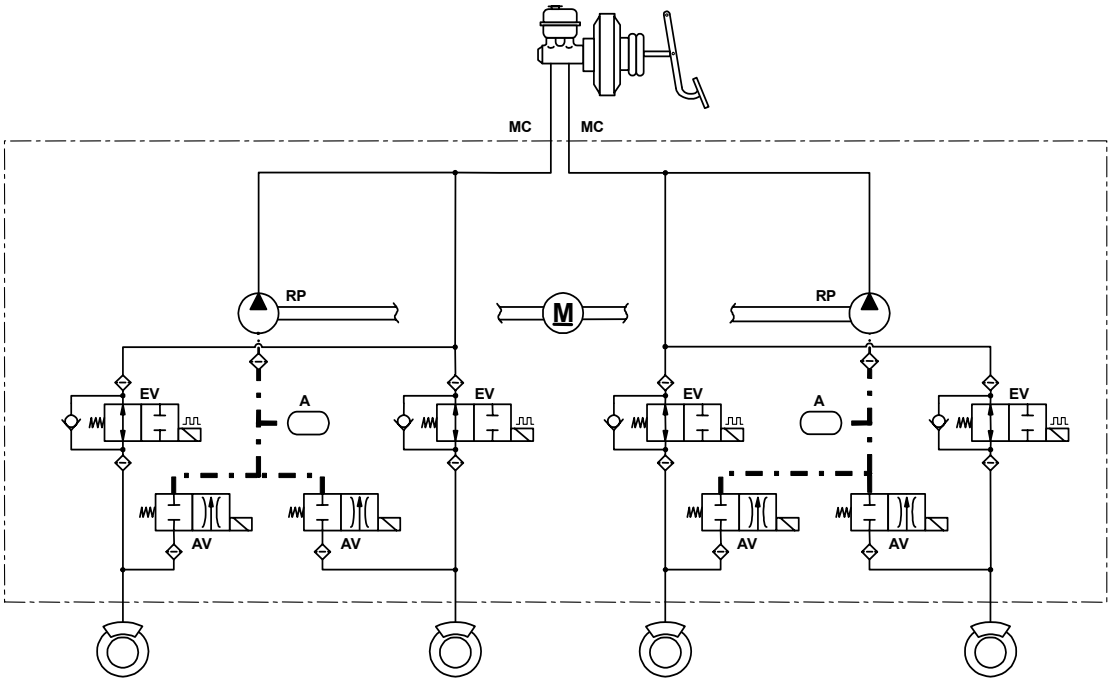
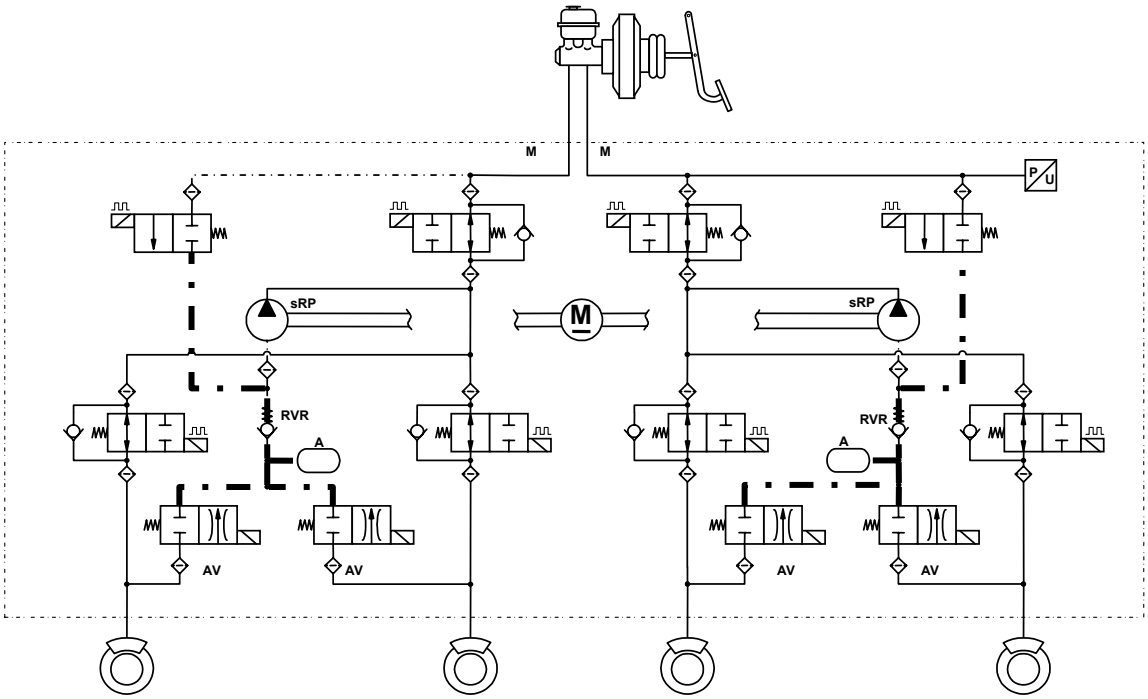
9 Attachments

9.1 Hydraulic Diagrams

9.1.1 ABS 9 (4-channel) – independent of the brake split configuration



9.1.2 ESP 9 – independent of the brake split configuration



The fat marked dotted lines within the ABS respectively ESP hydraulic unit shows the area of the secondary and priming circuit

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